

Unit and student details

Unit code	MEC3416	Unit title	Machine Design		
If this is a group assignment, each student must include their name and ID number and sign the student statement.					
Student ID	31444059	Surname	Abi Chebli	Given names	Jason
Student ID	31444660	Surname	Sherar	Given names	Reece
Student ID	29704693	Surname	Mavros	Given names	Alex
Student ID	32529457	Surname	Schmid Walter	Given names	Sean
	30588499		Morey		Claire

Assessment details

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Lecturer/tutor	Dr Jing Fu	Tutorial day and time	
Due date	23/10/2022	Date submitted	
Has any part of this assessment been previously submitted as part of another unit/course?			Yes ☐ No ☒

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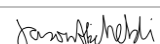
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Maintenance of Offshore Wind Turbines

Design Group 5

Jason Abi Chebli (31444059)
Alex Mavros (29704693)
Claire Morey (30588499)
Reece Sherar (31444660)
Sean Schmid Walter (32529457)

Final Report Submission for MEC3416 Engineering Design II, S2,
2022.

Executive Summary

This product is designed to provide versatile and large personal lift capabilities for servicing of offshore wind turbines whilst remaining non permanent and simple to use. The final design consists of 4 Major parts; The main platform, rack and pinion mechanism, scissor lift, and safety brace mechanism.

Important features of this design are the cargo and personnel platform which is big enough to allow for adequate cargo to be transported as well as comfortable enough for a service person to travel up and down the wind turbine. The rack and pinion mechanism is also an important part of the design that allows the lift to reliably and effectively transverse up and down the turbine in a time of 5 to 7 minutes.

Setting up the lift mechanism is simple as the operator can use the scissor lift to move and lift the platform into place on the racks of the wind turbine. To move up and down the turbine on the lift the operator can use buttons placed on a handle bar located at the top of the platform or with a remote control when just transporting cargo. The lift will run on 3 phase power from the boat used to get to the turbine.

Appropriate safety factors have been used throughout the critical components of the design. The component under the most stress is the output shafts of the motors which undergo 127.6MPa of stress.

The overall cost of the design is approximately \$16,093.85 and the total weight is 129.23kg.

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1 Introduction

Energy is essential to help humans live a comfortable life and as the population increases we require more energy. As a large portion of the energy generated comes from finite resources that once used will be gone forever, engineers have designed renewable energy resources to address this issue and make energy more sustainable. One particular renewable energy method used is a wind turbine which turns energy generated by the wind into electrical energy. Offshore wind turbines are typically very effective as offshore wind speeds are typically faster than on land, meaning more electrical energy can be generated. Servicing of wind turbines is crucial for their longevity, however, current methods such as landing on the turbine from a helicopter are typically unsafe and costly. Instead, arriving by boat and transporting oneself up from the base of the wind turbine is preferred. Therefore, the objective of the project is to design a hoisting system that will transport a servicing engineer and/or necessary equipment and materials up and down a 120 metre turbine tower in 5 to 7 minutes after arriving at the base of the tower via boat. Furthermore, the system created should be able to be easily transferred to and used on other wind turbines.

An innovative final product with different components was designed to achieve the objective. A rack and pinion system was chosen as the mechanical mechanism for transporting. The final design consists of a feasible and aesthetic platform that is able to secure and transport an engineer and any materials or products required up and down the wind turbine. This platform connects with a tower clamping system, which is designed to be fixed on all wind turbines and functions to secure the platform onto the tower, preventing it from falling. Furthermore, a scissor lift with wheels is incorporated in the final product to allow for the easy attachment, removal and transportation of the platform so that it could be used on other wind turbines. Finally, a unique safety brace mechanism is also created in this design to help catch the engineer in the very unlikely event of them falling off the platform. During the design of the final product, the weight, cost, feasibility and safety of the engineer was at the forefront of considerations.

Analysis was conducted to validate the feasibility and safety of critical parts and to optimise them for the final product. In particular, free body diagrams were used to perform hand calculations to determine whether the parts will fail and how this may occur. These hand calculations were supported by finite element analysis (FEA) done on the parts, which illustrates the distribution of stress that each part will experience. Furthermore, in some instances, FEA was used to help optimise the design of certain components to reduce unnecessary weight whilst also maintaining a safety factor of at least 3. Further investigation was also conducted into the serviceability and lifetime expectancy of components in order to make sure that breakdowns do not occur due to a lack of maintenance.

2 Product Overview

2.1 Preliminary Designs

In order to create the most viable final product, our design underwent three major preliminary design iterations before we concluded on the final design.

2.1.1 Version 1

Version 1 of our final design can be seen in Figure 1. Version 1 consisted of a dual rack and pinion system with the teeth of the racks facing towards each other. After brainstorming over eight different configurations of a rack and pinion system, this configuration was selected as we thought it would naturally make our system more stable than a single rack and pinion system and the inwards direction of teeth felt like an easier way to engage our pinions with the racks. Other major considerations made in this design include the use of a handlebar for the user to hold onto and control the platform. Additionally, a brake pedal is placed on the floor of the platform as a safety feature - whereby, the engineer's foot must be on it when transporting on the platform, otherwise, the platform will not move/stop. The side and back safety guide were also incorporated as an additional safety feature, to prevent the engineer from accidentally falling off the platform. Additionally, guide wheels were used to stabilise the platform on the racks.

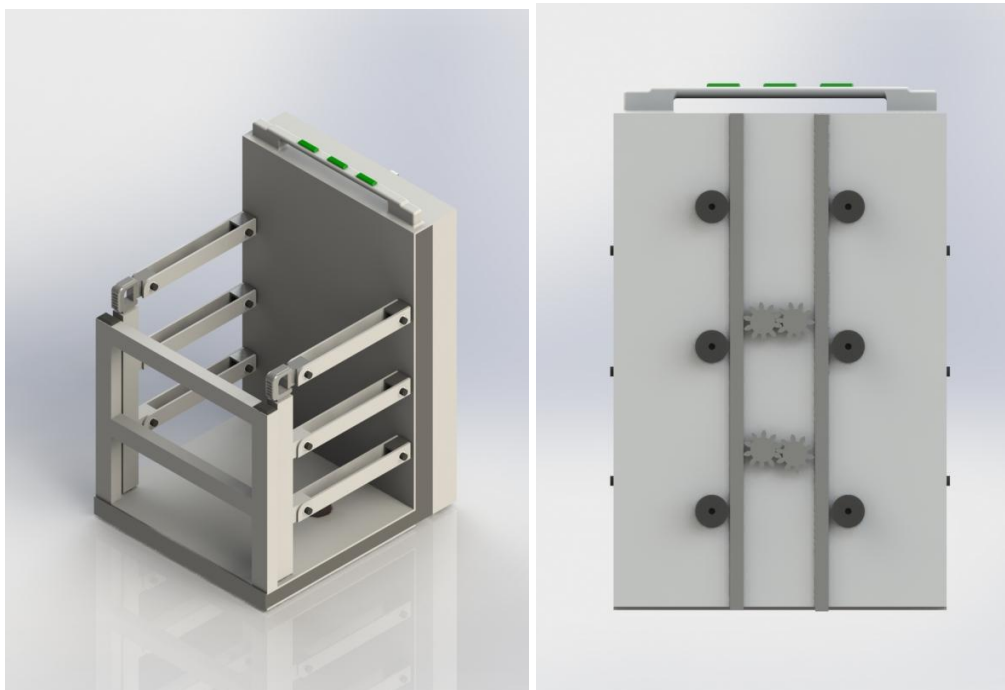


Figure 1: Platform V1.

2.1.2 Version 2

Whilst Version 1 had many highlights, it had some downsides. One critical drawback of the design was that there was nothing to prevent the system from tipping forward (due to a

moment caused by a human load). From this, version 2 was created - which can be seen in Figure 2. Version 2, now incorporated guide wheels that would not only support the rack (like version 1) but would also support the system from tipping forward. Additionally, the guide wheels designed were not generic wheels, but rather had a groove cutout of them, as it was thought that this combined with a custom rack would result in better stability. One other major improvement in this design was the incorporation of more controls - these included an on-off dial, a monitor to track statistics about the platform (such as elevation and speed), an emergency stop as well as an up and down button that would power a scissor lift that would be attached to the bottom of the platform (note that Figure 2 is missing the scissor lift that would attach, in theory to the bottom of the base of the platform). Furthermore, This design improved on the storage of equipment, as now there were at least three different storage compartments - one underneath, one near the bottom controls as well as one right next to the handle bar. Furthermore, the control bar, that has the necessary buttons to transport up and down, is now circular for better grip and comfort.



Figure 2: Platform V2.

2.1.3 Version 3

Version 2 was now a lot better functionally and as a result, we wanted to optimise the system which resulted in version 3, shown in Figure 3. The first improvement that could be made to version 2 was the height of the system. We desired the control bar to be at a height of around 1.3 metres from the base of the platform as we wanted to make it such that the average engineer can hold onto the bar with their forearms perpendicular to their body as we believed this to be the most comfortable for the engineer during the 7 minute trip. In version 2, the height was 1.5 metres which is too tall for the average engineer and so we made the platform height shorter. This was also done to reduce the weight of the system as we now wanted to make it lighter. Other changes were also made to reduce the weight, including the extra bars in the side and back safety guides, as these do not make much, if not any difference, to the safety of the engineer due to it being so low to the ground. Further changes also included removing some of the storage space, specially that located in the centre and

top of the platform as the shorter platform made it redundant and would take up more space. Furthermore, to reduce the weight of the overall platform, we also removed the idea of having a fixed scissor lift attached to the bottom of the system. Instead, we decided that the scissor lift will be an independent component that will stay on the ground as this would save weight. Moreover, we also moved the emergency button and the on-off dial to the top, as we believed it should be more easily accessible to the engineer that will be holding onto the control bar majority of the time. Furthermore, the monitor was also moved to the top as thought it would be impractical for the engineer to have to squat to read the monitor displays. We reduced the number of pinions from four to two as we found that the motor required is quite large and heavy and would also add to the expense of the project. The number of guide wheels were also reduced and the guide wheels were replaced with readily available wheels as we believed that making custom wheels with grooves cut in them would not only be expensive, but would make the cost of the rack way more expensive, as we would require 2 x 120 metres racks to be customised for each wind turbine. Overall, we were able to reduce the weight of the system from 210 kg in Version 2 to 132 kg in Version 3.

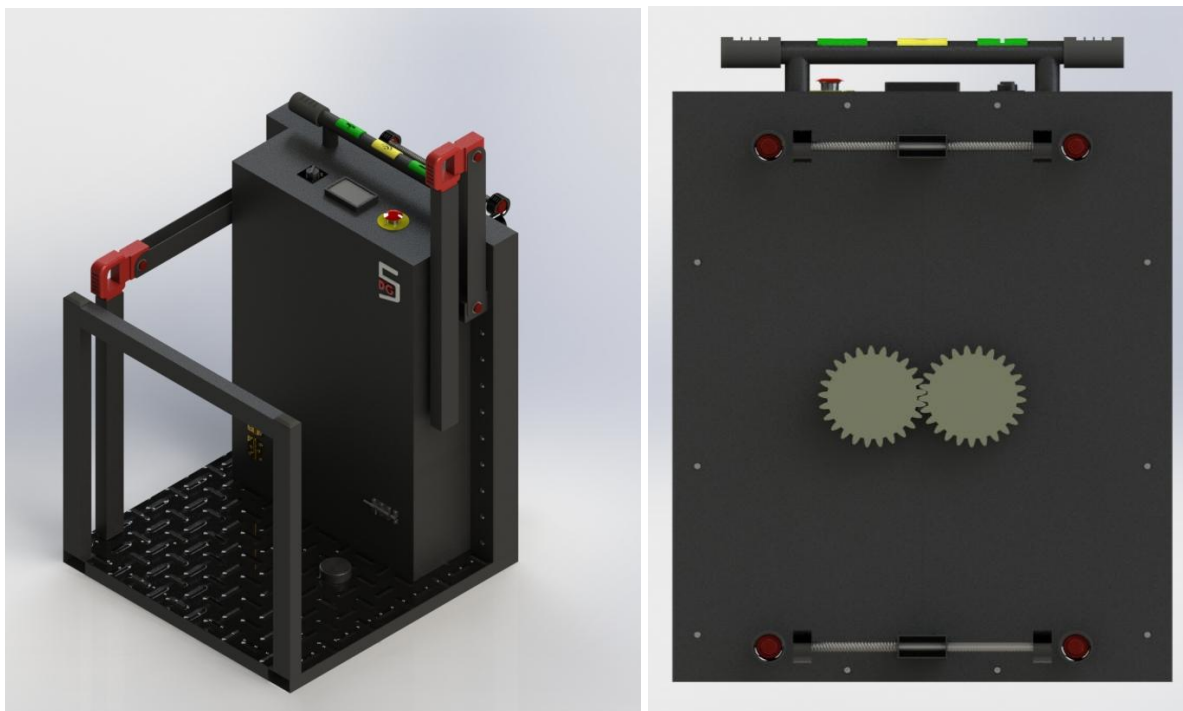


Figure 3: Platform V3.

While Version 3 was near perfect, it did have some flaws, specifically relating to the shaft of the motor. After performing a preliminary design analysis, we found that the shaft would not satisfy the safety factor requirement of 3. As such, the back compartment had to be redesigned.

2.2 Final Design

The final product design can be seen in Figure 4. All catalogues required can be found in Appendix B.1 and a cost breakdown of the platform can be found in Appendix C.2.1. There are four major components in this design, including the platform, safety brace, scissor lift and tower clamping system.

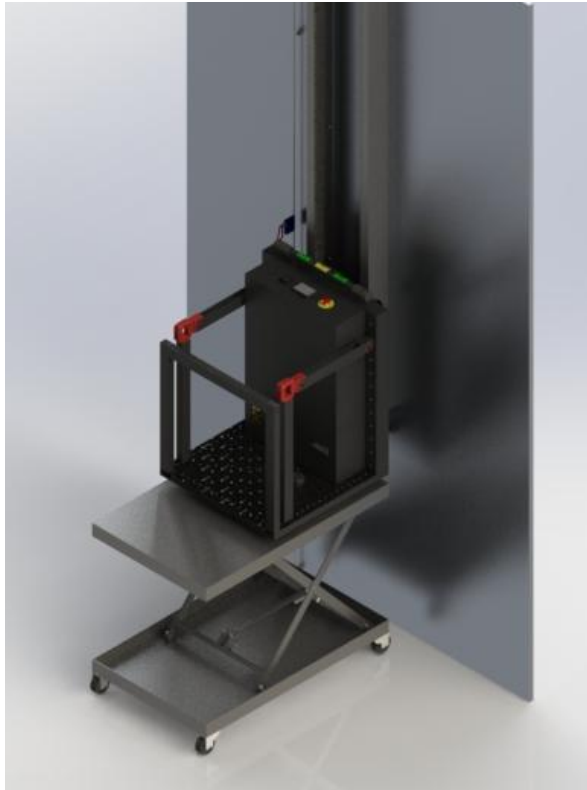


Figure 4: Final Design

2.2.1 Platform

The final iteration design of the platform can be seen in Figure 5.

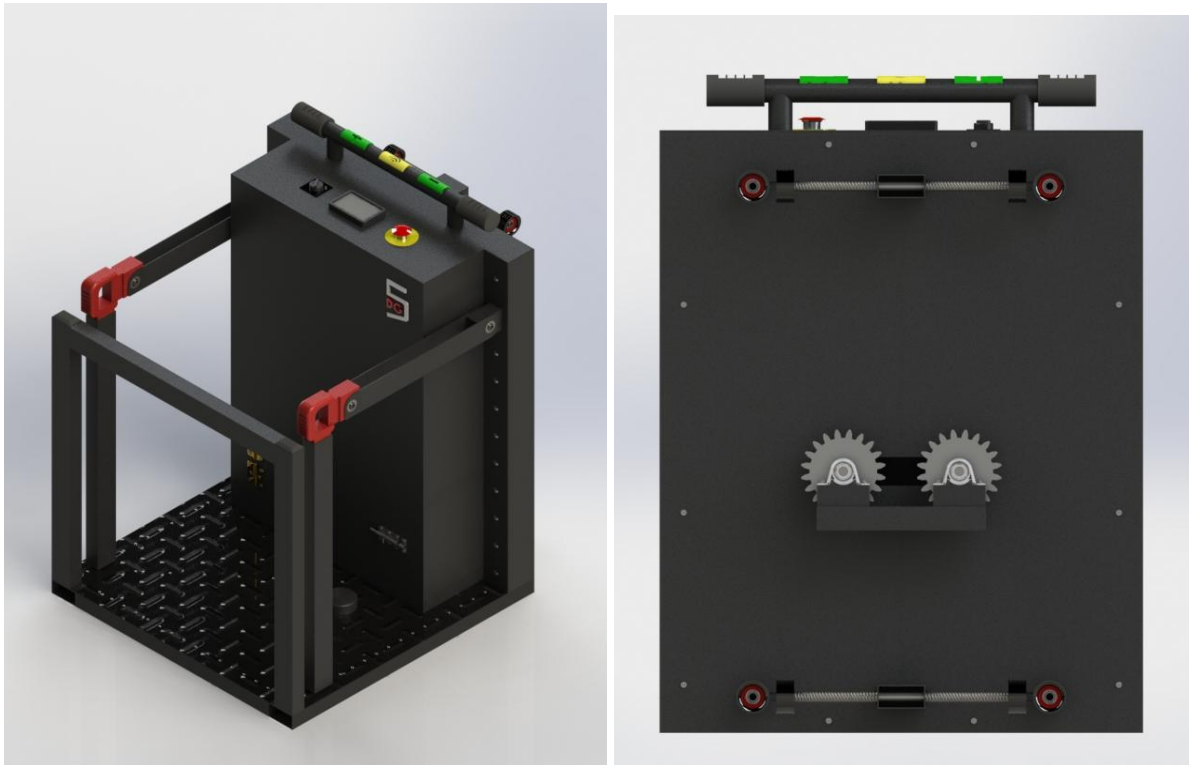


Figure 5: Final Design

The final design still has the dead man's switch pedal, which was first developed in version 1 of the platform, on the base. We kept this feature as we believed this safety feature is vital in the event that the engineer falls off the platform as the platform is incapable of travelling away from the engineer meaning that the engineer may be able to hopefully go back onto it. Additionally, the back safety guide and side safety guides, illustrated in Figure 6, remained the same as in version 3 as they will provide the engineer with safety from falling off the back and sides of the platform, respectively.

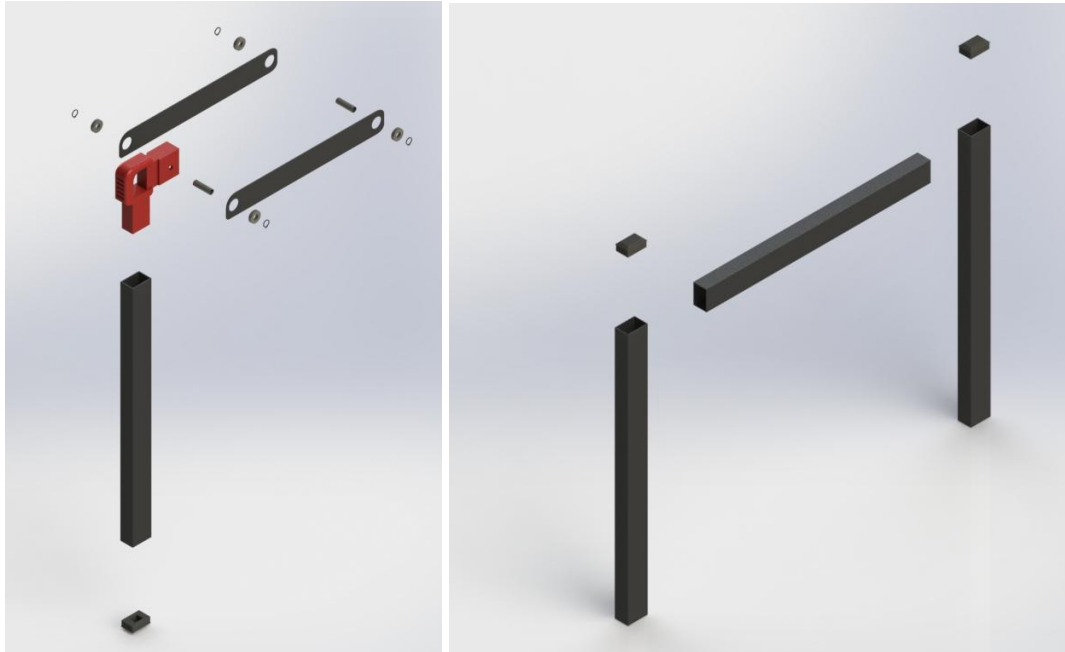


Figure 6: (a) Exploded view of the side safety guide. (b) Exploded view of the back safety guide.

Additionally, the side guides can easily be lifted up and folded when the engineer wants to get on and off the platform. Sensors will be used to make sure that the system cannot traverse up or down until the side arms are lowered as an added safety feature to prevent the engineer from accidentally falling.

As originally designed in version 2, the platform also has a storage compartment at the bottom of it where the battery can be placed alongside any equipment the engineer may need. A latch and hinge door is implemented to secure this compartment. This allows the engineer to safely store the equipment and is close to the wind turbine wall to reduce the moment it may cause. The storage compartment can be seen in Figure 7.



Figure 7: Platform storage compartment.

Similar to version 3, a control bar, shown in Figure 8 (a), allows the engineer to traverse up (left button), down (right button) or activate the remote control (middle button). Additionally, an on-off switch (left dial), an emergency stop (right button) and a monitor to track details of their position and speed (positioned in the centre) is placed before the control bar, as can be seen in Figure 8(a), so that it can be easily accessible and visible to the engineer. A proof of concept wireless remote control, illustrated in Figure 8(b), would be used to transport the platform without the engineer, such as in the event of sending up cargo.

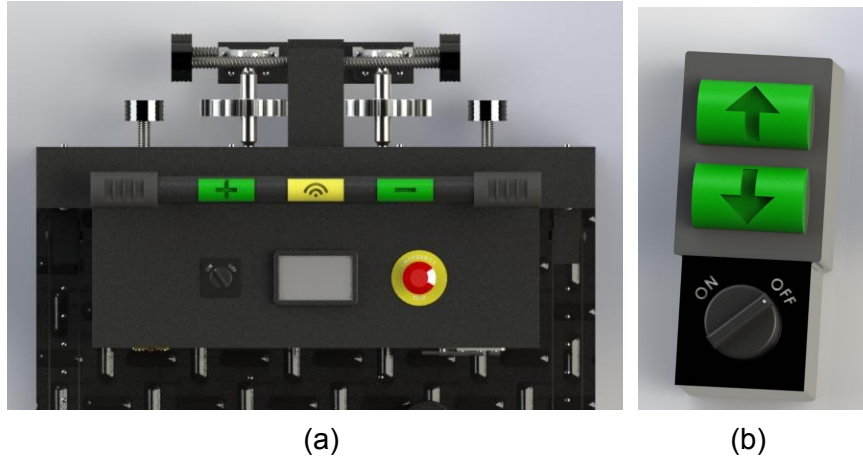


Figure 8: (a) Top controls. (b) Wireless remote control.

The back of the platform consists of two plates that slide in from the sides and are screwed on, as indicated in Figure 9. The design of those backplates allow for it to be easily removable without the need to remove any other components such as the guide wheels. This makes the motors, shafts and pinions easily accessible for maintenance.



Figure 9: Exploded back view of the platform.

Guide wheels are utilised to keep the platform steady. Each guide wheel, no matter its direction, is assembled in the same manner as the one illustrated Figure 10. Springs, bearings (interference fit within the wheel) and circlips are used to help fix the position of the guide wheels.

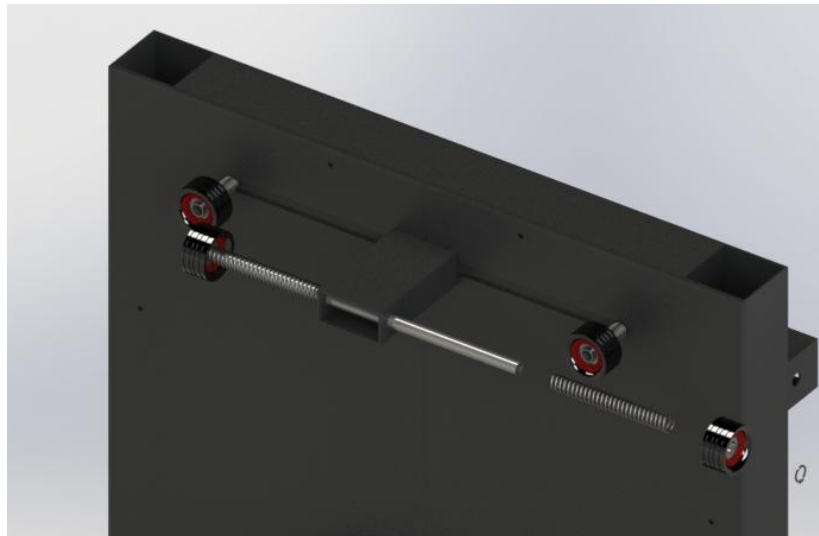


Figure 10: Exploded view of a guide wheel.

On the other hand, figure 11 illustrates how the top and bottom guide wheels keep the system perpendicular to the wall by applying forces in opposite directions onto a U-channel beam.

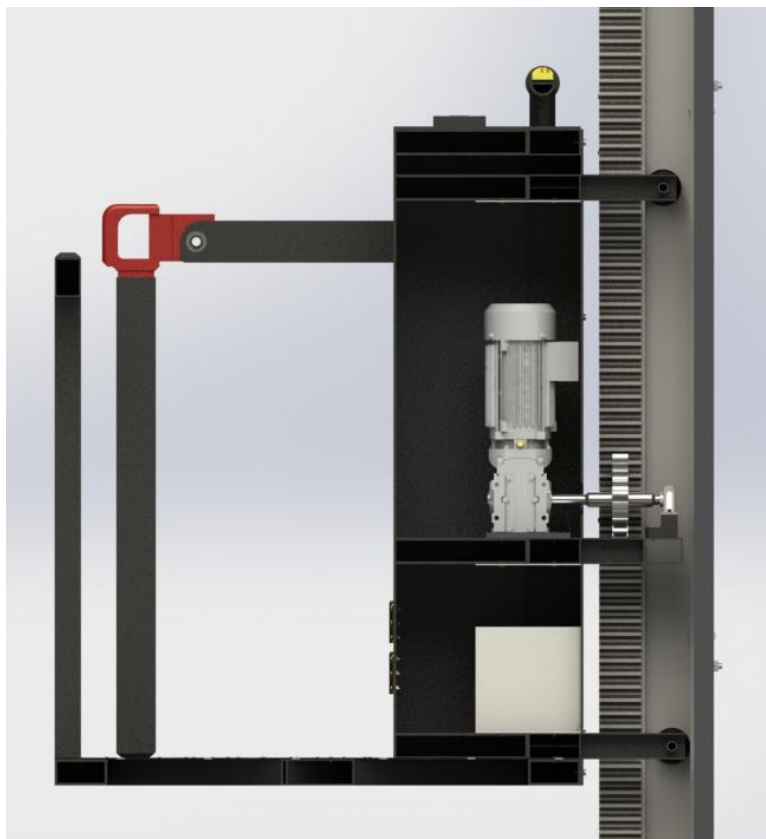


Figure 11: Section view illustrating the alignment of the platform to the wall.

To improve on the previous iteration flaw regarding the low safety factor for the shafts, as outlined in Section 2.1.3 Version 3, the motor and shaft component has been modified. Instead of using one larger motor that drives two pinions, we changed the design to contain two smaller motors that drive one pinion each. This not only increases the safety factor to an

appropriate level as will be outlined below, it also allows for the centre of mass of the platform to be centred, making calculations much better. Additionally, bearings were added to the otherside of the shaft, to help support the shaft of the pinions by helping distribute the loading. The two pinions will be driven by two 44.12 RPM gear reduced motors. This specific motor will use a helical worm gear to prevent the motor from back driving which is another added safety feature. The final motor and pinion components can be seen in Figure 12.

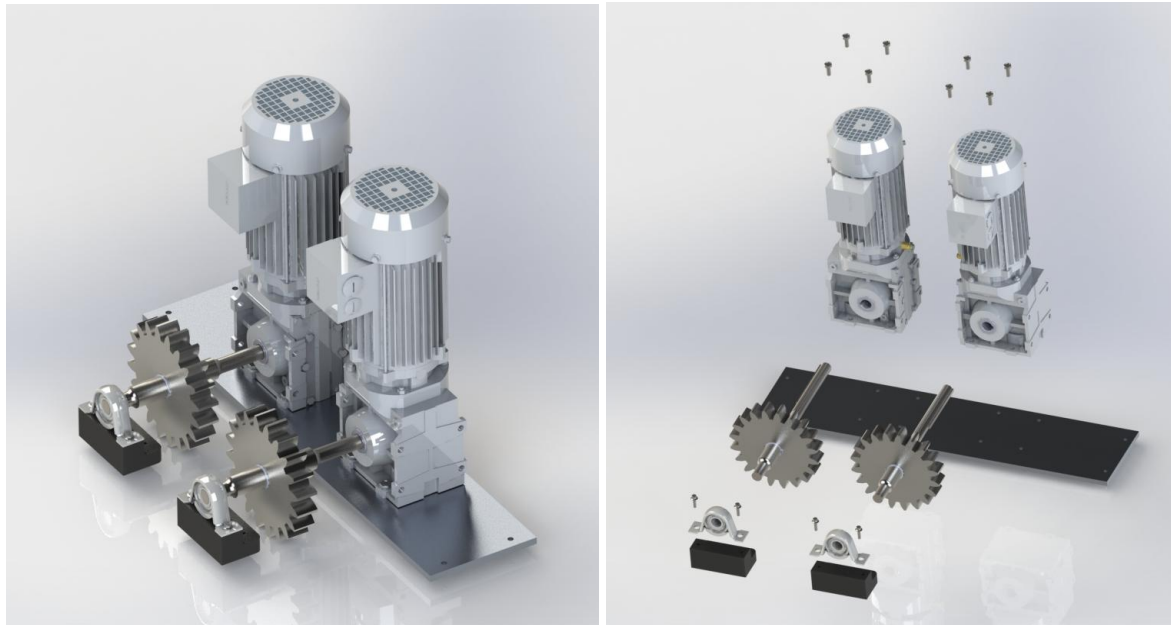


Figure 12: Isometric and exploded view of the motor and pinion system.

Four rectangular aluminium brackets (one being the one the motors attach to) are also incorporated in this design to help fix in place the front control dashboard to the chassis rectangular hollow sections (RHS). These are riveted in. These can be seen in Figure 13, which illustrates an exploded view of the platform. Further guidance regarding the assembly of the system can be found in Appendix A.

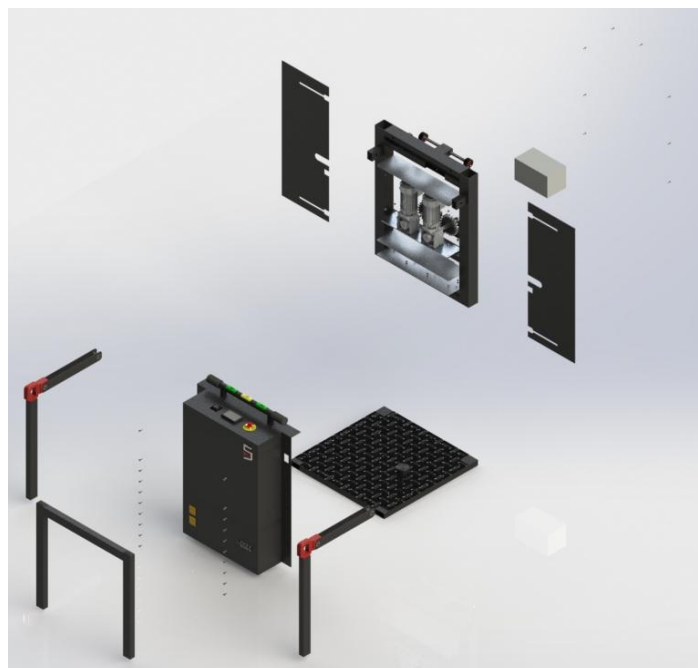


Figure 13: Exploded Back View of Platform.

2.2.2 Safety Brace

A customised safety brace, illustrated in Figure 14, was designed to be used in case of an emergency. All catalogues required can be found in Appendix B.2 and a cost breakdown of the safety brace can be found in Appendix C.2.2.

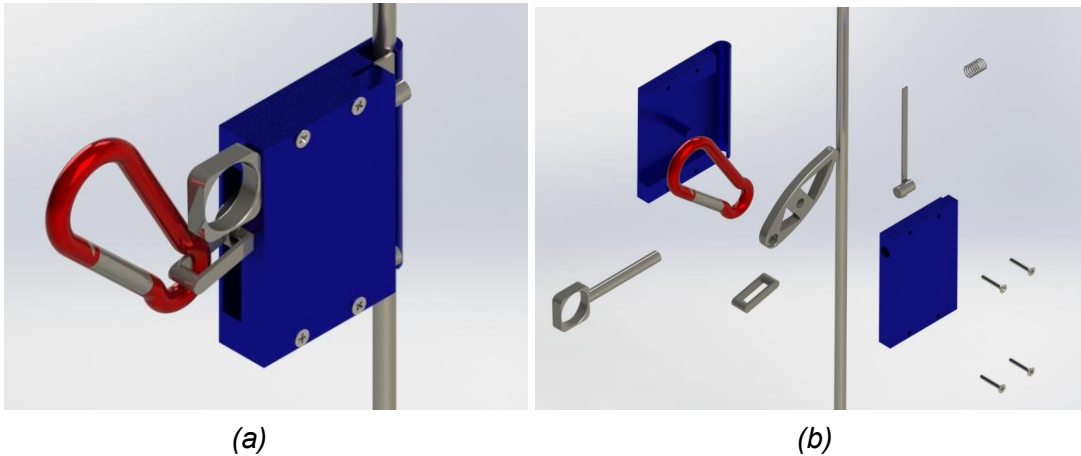


Figure 14: (a) Isometric view of Safety Brace. (b) Exploded view of Safety Brace.

A carabiner is used to attach the safety brace to a harness (that the engineer will need to wear). In the unlikely circumstance that the engineer falls, the carabiner will apply a downward force which will cause this brake pad to rotate around the centre pivot point. It will continue to rotate until the other end makes contact with the metal rod, at which point, the engineer will no longer be falling. Figure 15 illustrates this concept. The front cover of the safety brace was removed to make the mechanism more easily explained and understood.

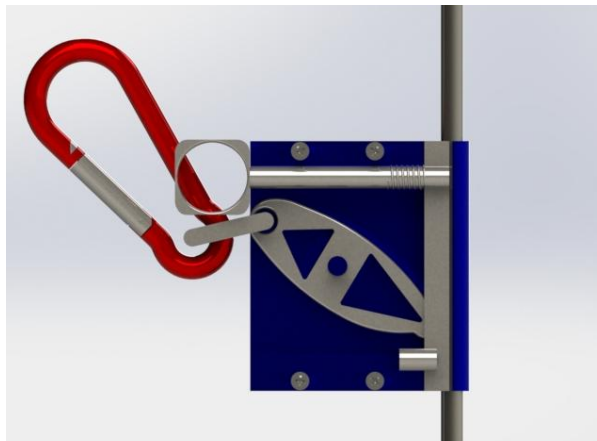


Figure 15: The brakepad of the safety brace catching onto the metal rod.

The safety brace was designed to be easily attachable and removable from the metal rod, when the engineer is not travelling up or down the wind turbine. To remove or place the safety brace onto the metal rod, the engineer must pull back a metal key with a finger in order for there to be enough space for the metal rod to slot through. Otherwise, since this key is attached to a slit of metal that blocks the rod from coming off, it will not attach or detach. Additionally, this mechanism is spring loaded, meaning the engineer needs to apply enough force to open the space, otherwise it will not come off. When the key is pulled back, it causes the spring to compress and so the spring will try to close the slot. The spring is incorporated into the design so that the engineer cannot be unintentionally unhooked. This concept is shown in Figure 16.

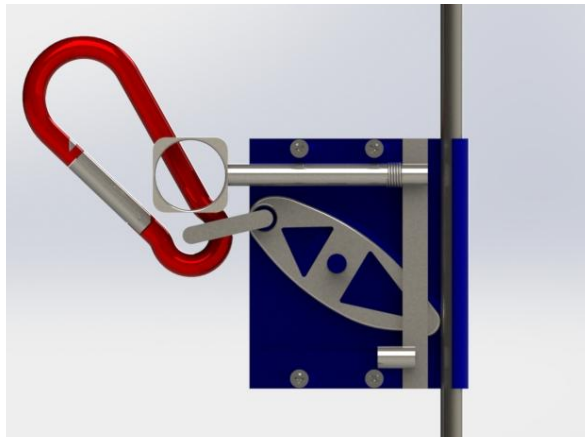


Figure 16: Attaching/detaching from the rod. As can be seen compared to the previous figure, when you pull the key back, it opens a slit for the rod to go through.

2.2.3 Scissor Lift

A scissor lift is required to transport the platform to the base of the wind turbine and to engage the pinion teeth with the rack teeth. As the platform weighs 129.23 kg, a scissor lift with wheels on it is required to allow for the engineer to transport the platform by themselves without having to strain themselves. Furthermore, these wheels can lock into place which prevents it from moving in the event of big gusts of wind. Additionally, the platform sits on top of the scissor lift and is not attached to the base of the platform to avoid making the platform too heavy.

A feasible scissor lift design can be seen in Figure 17. Guide wheels are attached to one end while the other end is bolted. With the other end pinned and this end a roller, this mechanism makes the scissor lift move up and down. Furthermore, a 12V, 500mm stroke, 3000N pull load electric linear actuator motor is incorporated in this design to lift the scissor lift up and down.

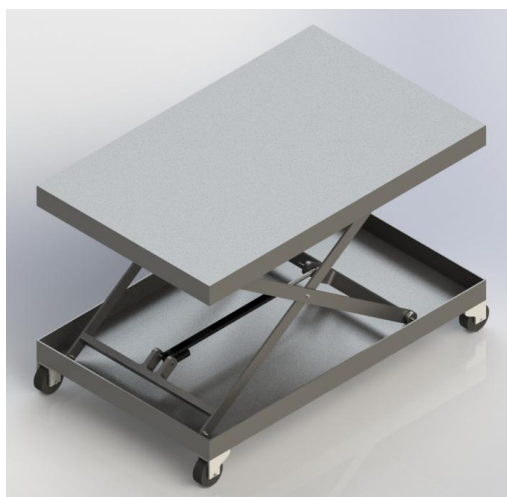


Figure 17: Isometric view of a feasible scissor lift design.

A proof of concept wireless remote control, illustrated in Figure 18 will be used to control the scissor lift - allowing it to move up and down.

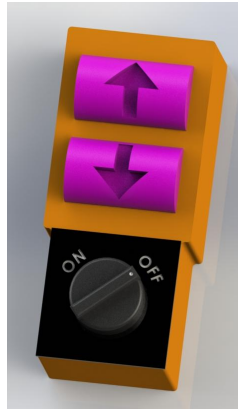


Figure 18: Scissor lift remote control.

While this design is feasible, more testing is required. While it would be valuable, due to the short timeframe for this project, we believe that the design of a scissor lift is out of the scope of this project as it could be a project on its own. As such, instead of making a scissor lift, for now, a scissor lift will be purchased, as discussed later on in Section 3.8, Appendix B.3 and Appendix C.2.3.

2.2.4 Tower Clamping System

As the platform uses a rack and pinion mechanism to transport the engineer up and down the wind turbine, the racks need to be fixed to the wall. To do so, U-channel beams are used to attach the rack to the wall. The U-channels face each other and provide the guide wheels a spot to sit in and a surface to act upon. The racks are module 8, which was chosen to match the pinions to allow for a smooth drive. Similar to the U-channels, the rack teeth face each other. Furthermore, a metal rod is attached to the side of the U-channels. The safety braces attach to this metal rod and use it, in the event the system fails or the engineer falls off the platform. Figure 19 illustrates the tower clamping system. All catalogues required can be found in Appendix B.4 and a cost breakdown of the tower clamping system can be found in Appendix C.2.4.



Figure 19: Close-up isometric view of the tower clamping system.

Rivets are used to secure the rack to the U-channel and to secure the U-channel to the wind turbine wall. Rivets are used over bolts as we do not expect the tower clamping system to be removed frequently unless for repair. An exploded view which shows how the system is assembled can be seen in Figure 20.

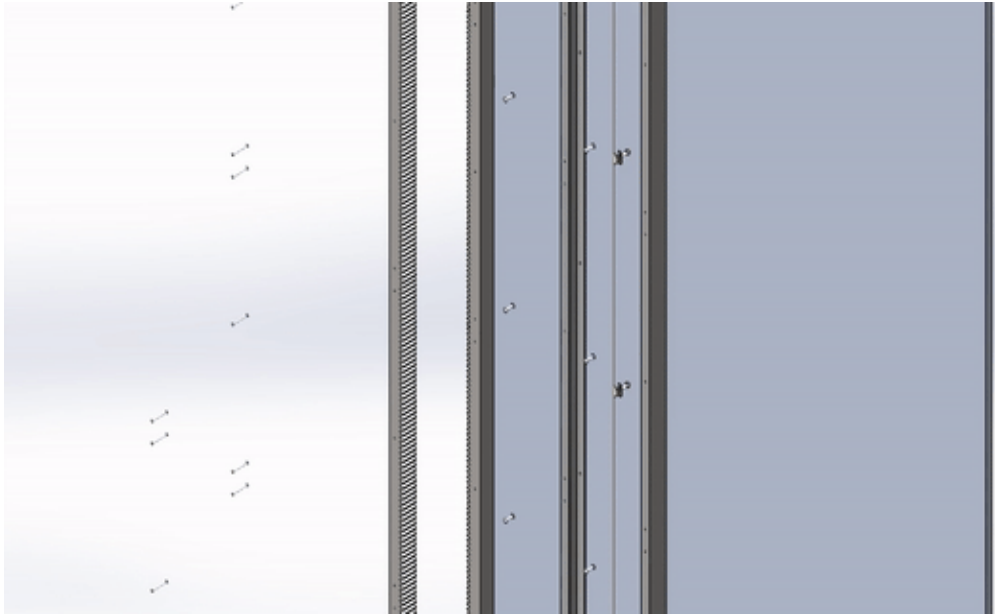


Figure 20: Exploded view of the wall.

3 Component Design and Selection

3.1 Platform Base

3.1.1 Hand Calculations

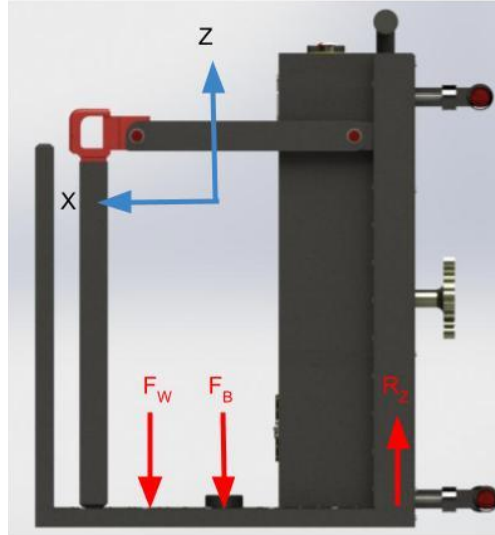


Figure 21: Free Body Diagram of the forces on the platform base

For these calculations it is assumed that:

- All loads are point loads
- Mass of person and cargo is 120kg
- The platform travels at a constant velocity
- The load and failure analysis is conducted in an ideal, frictionless situation
- No excessive noise or vibration
- Gravity is 9.81m/s^2

$$F_W = 49.656\text{kg} \times 9.81\text{m/s}^2 = 487.13\text{N} \quad F_B = 120\text{kg} \times 9.81\text{m/s}^2 = 1177.2\text{N}$$

$$\Sigma F_x = 0$$

$$\Sigma F_z = -F_W - F_B + R_z = 0$$

$$R_z = 1664.33\text{N}$$

Moment in beam taken at R_z

$$M_B = (487.13 \times 0.5) + (1177.2 \times 0.678)$$

$$M_B = 1041.70\text{Nm}$$

Shear Stress

$$\tau = \frac{F}{A} = \frac{1664.33}{(0.05 \times 0.1) - (0.044 \times 0.094)}$$

$$\tau = 1.93\text{MPa}$$

Bending Moment

$$\sigma_B = \frac{Mc}{I} \quad I = \frac{bd^3}{12} - \frac{b_i d_i^3}{12}$$

$$I = 3.7439 \times 10^{-7}$$

$$\sigma_B = \frac{1041.70Nm \times 0.025}{3.7439 \times 10^{-7}}$$

$$\sigma_B = 69559817.3Pa$$

$$\sigma_B = 69.56MPa$$

Max stress from bending stress

$$S_y = 448.0025MPa \text{ Tensile Strength for Cast Steel}$$

Safety Factor

$$S.F = \frac{448.0025}{69.56} = 6.44$$

3.1.2 FEA

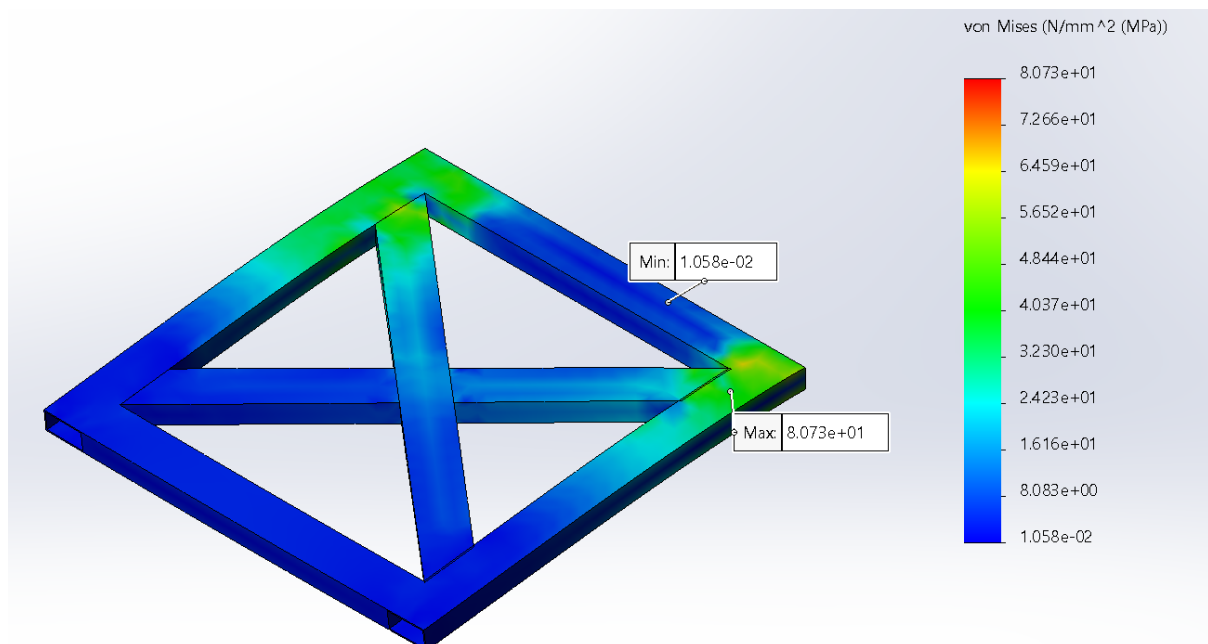


Figure 22: FEA of platform base

From this FEA analysis in Figure 22 we can see that the base platform will have a safety factor of 5.55 Which is still acceptable with the max stress in the base being 80.73MPa which is higher than calculated by hand due to some assumptions made for the hand calculations such as, treating the loads as point loads when they are better modelled as distributed loads. The top plate was excluded from this FEA as it does not add much to the overall structure and instead acts to distribute the load over the structure which can easily be represented in the FEA without the need for the top plate.

3.1.3 Servicing

In theory the base platform should not need to be replaced. The main servicing for this part will be to ensure it is still structurally sound. If the base is found to not be structurally sound then it will need to be replaced or fixed if possible before use. Refer to Appendix B.1 and C.2.1 for parts and cost of platform.

3.2 Base Welds

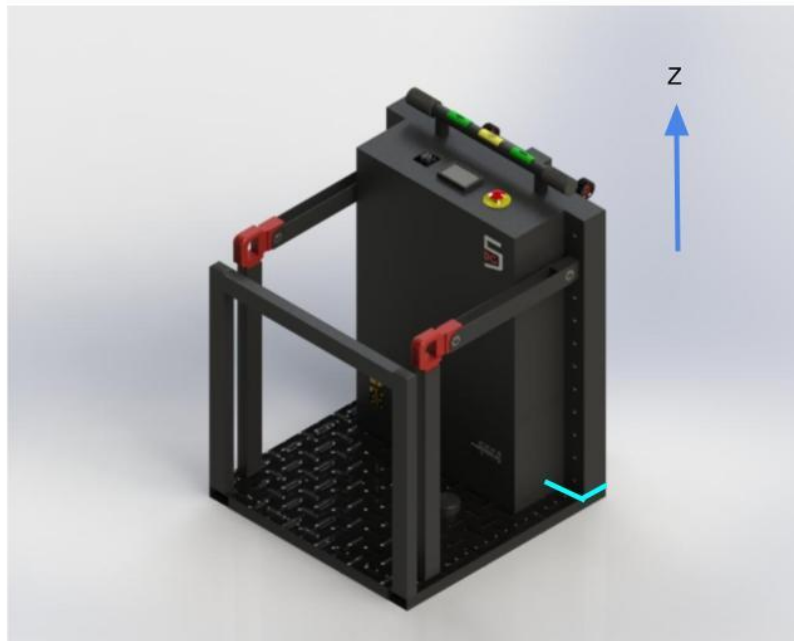


Figure 23: Diagram of base welds on platform image

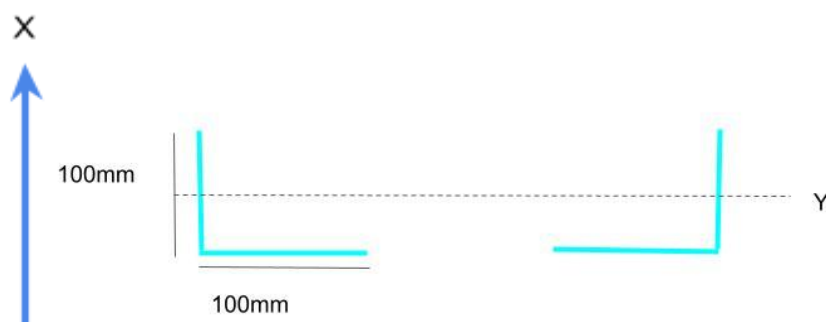


Figure 24: Detail view of weld not drawn to scale

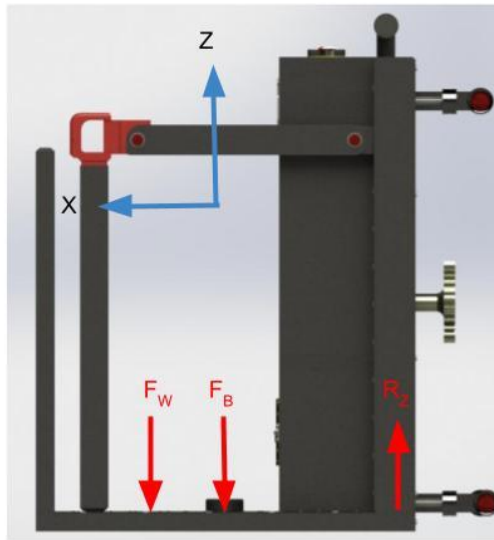


Figure 25: Free body diagram of forces on base that will be applied to the weld

Using E60 Welds since this will give a relatively high yield stress of 345MPa meaning the welds can have a higher safety factor than is required with little extra expense.
Weld is reflected about the X axis

Calculating the Vertical and horizontal moments of inertia

$$I_v = \frac{100^3 t}{12} = 83333.33 tmm^4$$

$$I_h = 200t\left(\frac{100}{2}\right)^2 = 500000 tmm^4$$

Combining the moments of inertia gives moment of inertia about Y

$$I_y = 2 \times I_v + I_h = 666666.67 tmm^4$$

Calculating Bending Stress about the Y-Y axis. Using Bending moments from the platform base calculations converting to Nmm

$$\sigma = \frac{243562.68 \times 50mm}{1166666.67t} + \frac{798141.60 \times 50mm}{666666.67t} = \frac{78.1278}{t} MPa$$

Calculating Shear Stress About the Y-Y axis

$$\tau = \frac{1664.33}{(200+200)t} = \frac{4.1608}{t} MPa$$

Resultant Stress

$$\tau_{resultant} = \sqrt{\frac{78.1278^2}{t^2} + \frac{4.1608^2}{t^2}} = \frac{78.2385}{t} MPa$$

Calculating t using a safety factor of 6

$$\frac{78.2385}{t} = \frac{345(0.58)}{6}$$

$$t = 2.35mm$$

Therefore h is

$$h = \sqrt{2}t = 3.32mm$$

These weld sizes are feasible as the thickness of the material that is being welded to is 3mm. Refer to Appendix C for cost of welds.

3.2.2 Servicing

Servicing of this weld would require a visual check every time an operator sets up the lift for use to ensure no visible changes have happened since last being used. Then annually the weld can be checked with a fillet gauge to ensure it is still inspected considering the environment the lift is in.

3.3 Gearbox Output Shafts

3.3.1 Motor Calculations and Selection

Driving the shafts requires a power input from two separate motors. The motor selection requires for the motor to supply sufficient power at an RPM range specified by the design brief. The torque provided must also be sufficient to drive the pinion against the force experienced by the pinion teeth, note that by defining 2 of these 3 specifications, the other can be calculated. Due to the symmetry of the platform the motor specifications will be identical, as the pinion forces are identical, but will drive each shaft in opposite directions.

The platform is required to move the platform 120 metres in 5 - 7 minutes, carrying a load of up to 100kgs.

$$m_{platform\ and\ load} \approx 250\ kg$$

$$v_{7min} = \frac{120}{7(60)} = 0.29\ m/s$$

$$v_{5min} = \frac{120}{5(60)} = 0.4\ m/s$$

$$P_{platform} = Fv = mgv$$

For minimum power, the slowest platform velocity will be used:

$$P_{platform} = 250(9.81)(0.29) = 0.699\ kW$$

This power is delivered by two motors:

$$P_{motor} = \frac{P_{platform}}{2} = 0.349\ kW$$

$$P_{motor} = \tau\omega, \quad \omega = \frac{v}{r} = \frac{0.29}{0.08} = 3.625\ rad/s = 34.6\ rpm$$

$$\tau = \frac{P_{motor}}{\omega} = \frac{349}{3.625} = 96.3\ Nm$$

From the hand calculations above the minimum specifications for the motors have been found. Using these values, an appropriate motor can be found using Siemens SIMOGEAR online configuration tool [1]. The motors selected were the SIMOGEAR CA29-LE80MH4P geared motor. These configurations contained the motor and a correct gearbox allowing for appropriate speed and torque.

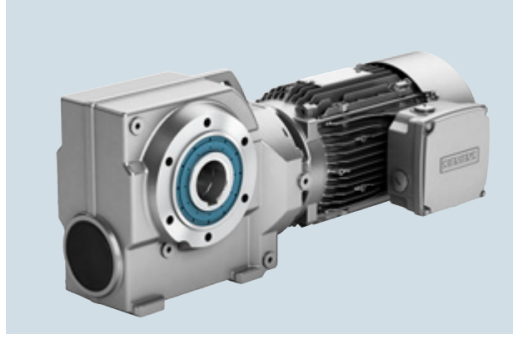


Figure 26: Siemens SIMOGEAR helical worm geared motor [1].

The selected motor contained a helical worm gearbox, stopping backdrive when the motor is switched off. The power of the motor is 0.55 kW, providing 107 Nm of torque at 44.12 rpm. Due to inefficiencies in the gearbox, no 0.37 kW geared motors provided ample torque and speed. The increased speed of 44.12 rpm is still less than the 5 minute travel time outlined previously (equivalent to 47.7 rpm) and thus still matches the design specifications. The selected configuration is also a 3 phase motor, allowing for a smoother application of power leading to smoother operation and less fatigue. The power is supplied using leads from a generator located at the bottom of the wind turbine platform from where the mechanism is mounted.

3.3.2 Hand Calculations

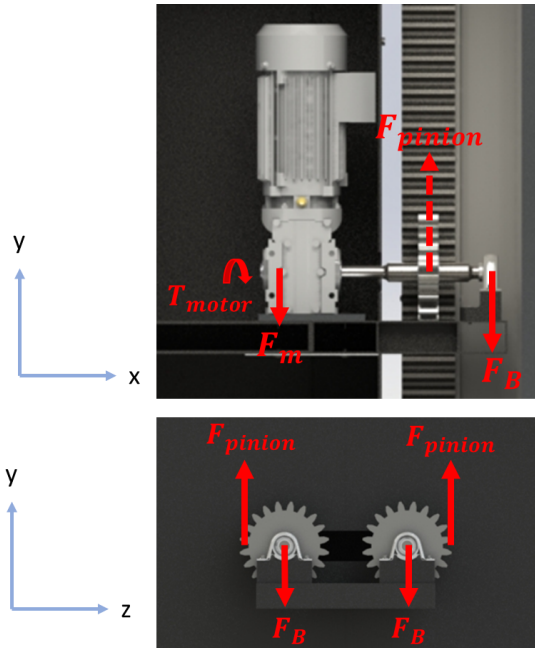


Figure 27: Free body diagram of motor shafts

The since the platform is symmetrical, the forces acting on each shaft are identical, therefore only one shaft needs to be analysed:

$$\sum F_x = 0 \text{ -- no axial forces}$$

$$F_{pinion} = 1222.4735 \text{ N}$$

$$\sum M_m = 0 = l_{pinion}F_{pinion} - l_{bearing}F_B$$

$$0.2488(1222.4735) = 0.33655F_B$$

$$F_B = 320.193 \text{ N}$$

$$\sum F_y = 0 = F_{pinion} - F_m - F_B$$

$$F_m = 1222.4735 - 320.193$$

$$F_m = 320.193 \text{ N}$$

$$\sum M_{axial} = 0 = T_{motor} - r_{pinion}F_{pinion}$$

$$T_{motor} = (0.08)(1222.4735) = 97.798$$

Calculating maximum bending moment and maximum torsion:

$$M_{max} = l_{pinion} F_m = 0.2488(320.193) = 79.55 \text{ Nm}$$

$$T_{max} = 97.798 \text{ Nm}$$

The initial shaft was designed using the ASME standard for shaft design. This was done by application of the following formula:

$$d = \sqrt[3]{\frac{32n_s}{\pi} \sqrt{\left(\frac{M_{max}}{\sigma_e}\right)^2 + \frac{3}{4}\left(\frac{T}{\sigma_y}\right)^2}}$$

Where, n_s = safety factor, σ_e = endurance limit, σ_y = yield strength

Using steel alloy: $\sigma_{UTS} = 723.83 \text{ MPa}$, $\sigma_y = 620.44 \text{ MPa}$

$$\sigma_e = S_n' C_L C_G C_S C_T C_R$$

Assuming 90% reliability, operating at normal temps, 20-30mm diameter shaft, surface roughness of 0.74 and a SF of 3.

$$\sigma_e = 0.5(723.83)(1)(0.9)(0.74)(1)(1) = 241.0339 \text{ MPa}$$

$$d = \sqrt[3]{\frac{32(3)}{\pi} \sqrt{\left(\frac{79.55}{(241.0339 \times 10^6)}\right)^2 + \frac{3}{4}\left(\frac{91.798}{620.44 \times 10^6}\right)^2}}$$

$$d = 22.18 \text{ mm}$$

Since the motor has a maximum shaft diameter of 20mm, the shaft was designed to have two steps, one to 25mm after the motor and one to 30mm near the pinion to decrease the deflection.

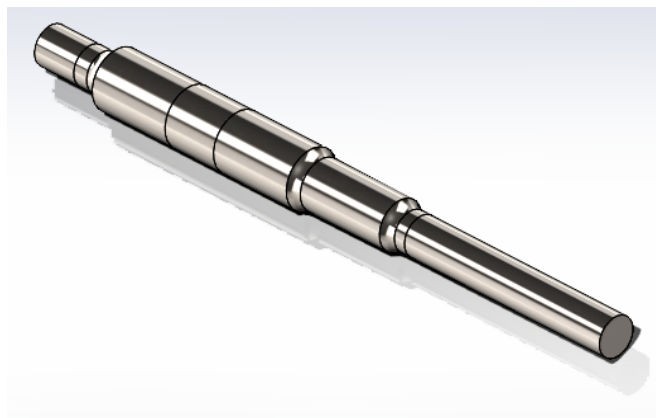


Figure 28: Initial shaft design.

To calculate maximum stress, stress values at each fillet must be found. The following calculations show the calculations for the fillet at which the maximum stress occurred:

$$K_{t \text{ bend, fillet } 1} = 1.29, K_{t \text{ torsion, fillet } 1} = 1.11$$

$$M_{fillet\ 1} = l_{fillet\ 1} F_m = 0.13011(320.193) = 41.66\ Nm$$

$$\sigma_{fillet\ 1} = \frac{My}{I} = \frac{32M}{\pi d^3} = \frac{32(41.66)}{\pi(0.02^3)} = 68.425\ MPa$$

$$\tau_{fillet\ 1} = \frac{16T}{\pi d^3} K_t = \frac{16(97.798)}{\pi(0.02^3)} (1.09) = 67.86\ MPa$$

$$\tau_{max} = \sqrt{\tau^2 + \left(\frac{\sigma_a + \sigma_b}{2}\right)^2}, \sigma_a = 0$$

$$\tau_{max} = \sqrt{67.86^2 + \left(\frac{68.425}{2}\right)^2} = 76\ MPa$$

$$SF = \frac{0.58\sigma_y}{\tau_{max}} = \frac{0.58(620.44)}{76} = 4.73$$

To finish the shaft design, splines were added to ensure complete power transmission between the pinion and shaft. A keyway was also added to the shaft as per the Siemens SIMOGEAR installation guide.



Figure 29: Initial completed shaft design

3.3.3 FEA

To model the keyed shaft motor attachment, a fixed geometry was used on the face of the keyway in contact with the driving side of the motor. To model the bearing, a bearing fixture was used, allowing for rotation without any axial or radial translations. Upon the first simulations, it appeared that a FEA singularity was appearing on the edge where the key meets the shaft.

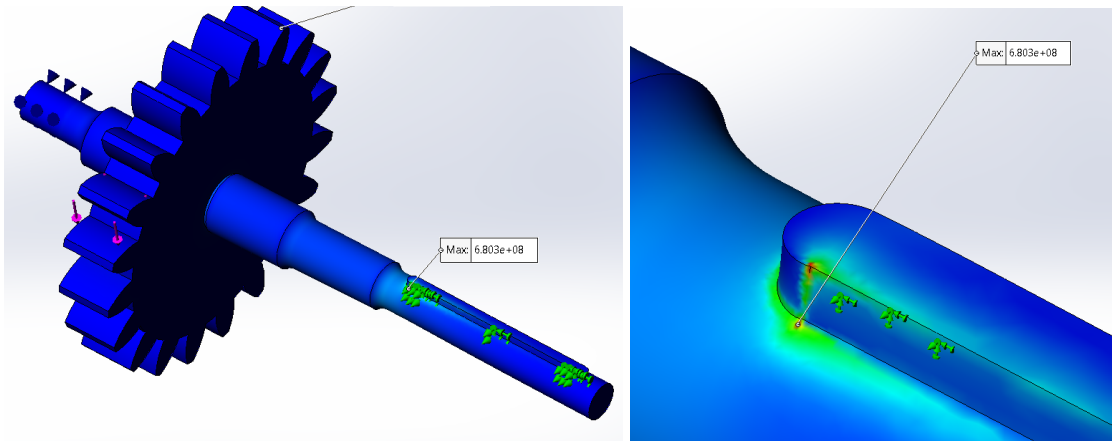


Figure 30: Initial FEA analysis. Left image displays the FEA fixtures and loads applied. Right image shows the singularity, where there is sharp points of extremely high stress (image shows close to 700MPa)

Known to occur at sharp edges, singularities create stresses that approach infinity as the mesh gets more refined, thus giving an unrealistic approximation of the stresses at that area. A convergence test over 5 iterations with decreasing mesh size was conducted to verify this, and its results displayed singularity characteristics. Since the key and keyway were designed based on the motor assembly guide, we believe it is safe to assume that stress concentrations estimated by the model are unrealistic, and for the purpose of shaft analysis can be ignored.

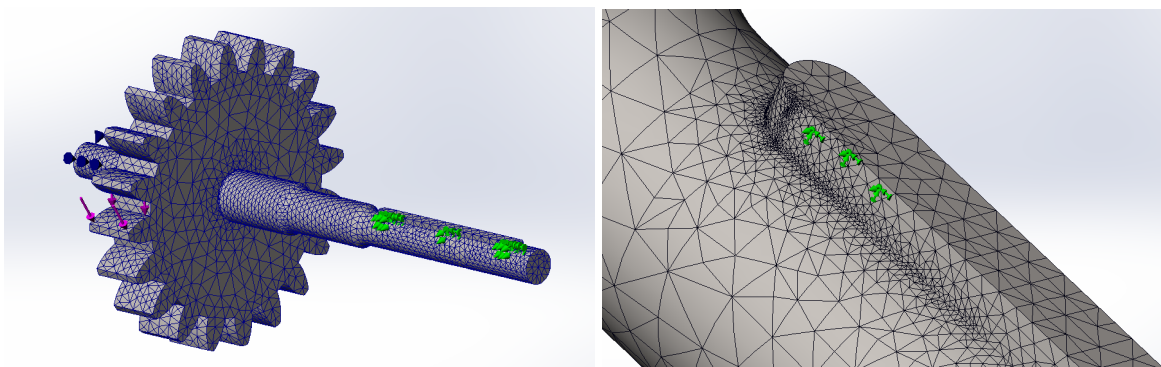


Figure 31: Mesh changes through 5 interactions from h-adaptive solver. The image on the left shows the initial mesh, while the image on the right shows the adapted mesh after 5 interactions. Note the higher concentration of nodes where the singularity occurs.

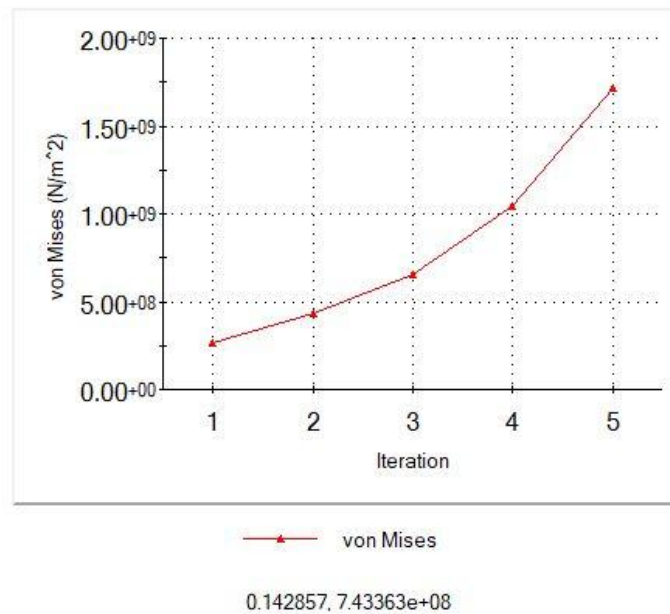


Figure 32: Graph of maximum von Mises stress plotted against iteration number. As iteration number increases and mesh refines, the maximum stress continues to increase, indicating a singularity and not indicative of real world behaviour.

To continue the analysis, the key and keyways were removed. The motor attachment was then modelled using a fixed geometry feature at the end of the shaft and an advanced fixture limiting radial translation on the cylindrical face which would be in contact with the collar of the motor. This combination modelled the transition fit present between motor collar and shaft. The fixed geometry was required as the static model required a reaction force to stabilise shaft rotation to limit excessive displacement.

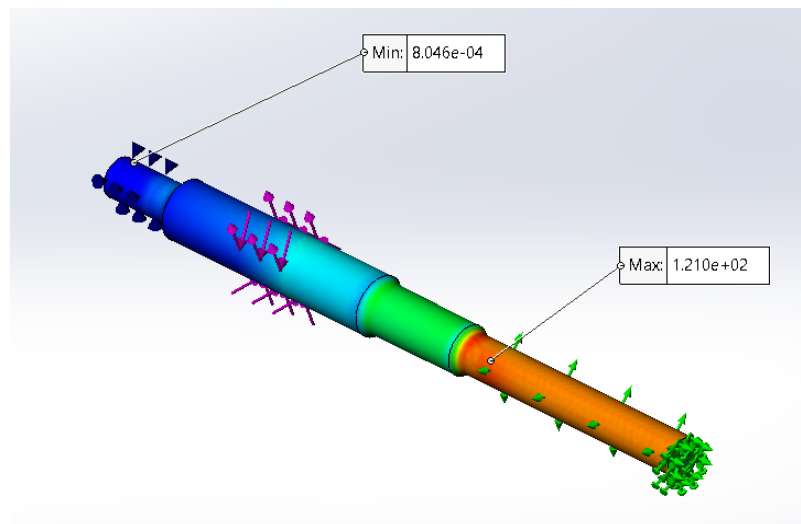


Figure 33: Initial shaft optimisation simulations

Optimization of the design is done to minimise the weight of the shaft, thus reducing material costs. The dimensions being altered were the diameter of the shaft where the pinion was attached and the diameter that was added as estimated by the ASME calculations. The pinion shaft radius was allowed to decrease from 15mm (initial) to 12.5mm in steps of 0.5, while the ASME shaft radius ranged from 12.5mm to 10.5mm in steps of 0.5mm. The

constraints of the system were its safety factor, which we required to be greater than 5, and deformation, which could not exceed 0.25mm. These benchmarks were chosen as the shaft is a critical load bearing component, and thus a higher safety factor than minimum of 3 specified in the design specifications was more suitable. Deformation must be kept low, as any deflection will affect the pinions interaction with the rack, which may influence efficiency of the system and create vibrations and rough operating conditions leading to a reduction in the lifetime of many components (bearings, motor, the shaft itself, fasteners, ect.) For the optimization process, the shaft had to be analysed separately from the pinon. The forces on the pinion were modelled on the shaft as the 1,222 N force acting downwards on the centre of the shaft, with the torque load being added on the same section of the shaft. The splines and circlip grooves, along with the keyway were also removed from the to simplify the optimization model, as these areas were known not to be locations of maximal stress. The 10mm fillets on edges where the shaft changed diameter were also removed, as the dimensions that we were changing did not always produce an edge, and thus caused an error.

	Scenario number	Pinion Shaft Radius (mm)	ASME Shaft Radius (mm)	Minimum Factor of Safety	Displacement (mm)	Mass (g)
Initial Value		15	12.5	4.706	0.2178	1282.66
Optimal Value		12.5	10.5	5.091	0.2369	1021.89
	1	12.5	10.5	5.091	0.2369	1021.89
	2	13	10.5	5.035	0.2368	1058.9
	3	13.5	10.5	5.122	0.2377	1097.36
	4	14	10.5	5.13	0.2395	1137.27
	5	14.5	10.5	5.149	0.242	1178.64
	6	15	10.5	5.089	0.2452	1221.46
	7	12.5	11	4.838	0.2291	1036.19
	8	13	11	4.902	0.2287	1073.2
	9	13.5	11	4.843	0.2292	1111.66
	10	14	11	4.889	0.2307	1151.58
	11	14.5	11	4.763	0.233	1192.94
	12	15	11	4.935	0.2358	1235.76
	13	12.5	11.5	4.71	0.223	1051.16
	14	13	11.5	4.769	0.2223	1088.17
	15	13.5	11.5	4.834	0.2226	1126.63
	16	14	11.5	4.78	0.2239	1166.55
	17	14.5	11.5	4.823	0.2258	1207.91
	18	15	11.5	4.869	0.2285	1250.73
	19	12.5	12	4.657	0.2182	1066.79
	20	13	12	4.635	0.2173	1103.8
	21	13.5	12	4.764	0.2174	1142.26
	22	14	12	4.718	0.2184	1182.18
	23	14.5	12	4.777	0.2202	1223.54
	24	15	12	4.708	0.2225	1266.36
	25	12.5	12.5	4.708	0.2225	1266.36
	26	13	12.5	4.653	0.2133	1120.1
	27	13.5	12.5	4.68	0.2132	1158.56
	28	14	12.5	4.648	0.214	1198.48
	29	14.5	12.5	4.568	0.2156	1239.84
	30	15	12.5	4.706	0.2178	1282.66

Figure 34: Shaft optimisation scenarios and results

The optimised values correspond to the smallest diameters, which is counterintuitive. However, this is caused by a reduction in the magnitude of stress concentrations which occur when a stepped shaft changes diameter. In the final shaft design, the ASME shaft diameter was removed completely, as the small step from 20 mm diameter to 21mm diameter was not worth the benefit of the added stiffness compared to the stress it added. Fillets, keyways, circlip grooves and splines were readded in the final design. Ultimately the optimised shaft resulted in an approximate 20% weight reduction, while slightly increasing its SF without excessive deflection

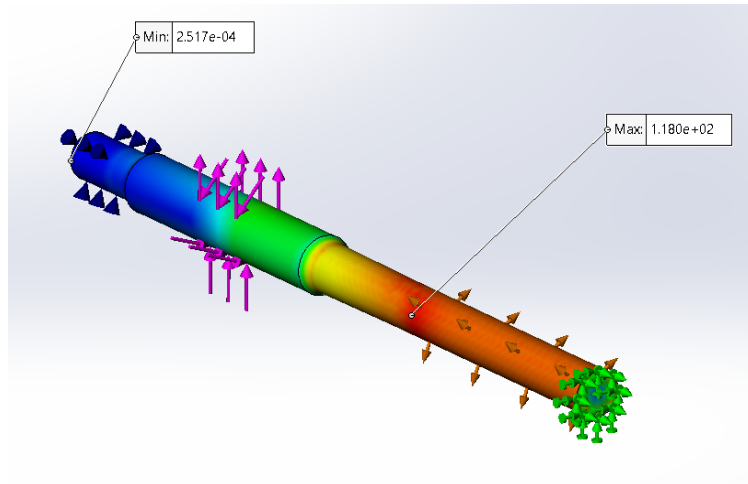


Figure 35: Optimal shaft design simulation

Analysis of the optimised shaft with the pinion attached was conducted, and the maximum stress occurred at the circlip grooves, with at SF of 4.9. This indicates that it may have been a limiting factor if it was present during the optimization process. However, since the SF was only reduced by 0.1 from the estimate was small it was deemed negligible. A detailed drawing of the final shaft design is provided in Appendix A.2.

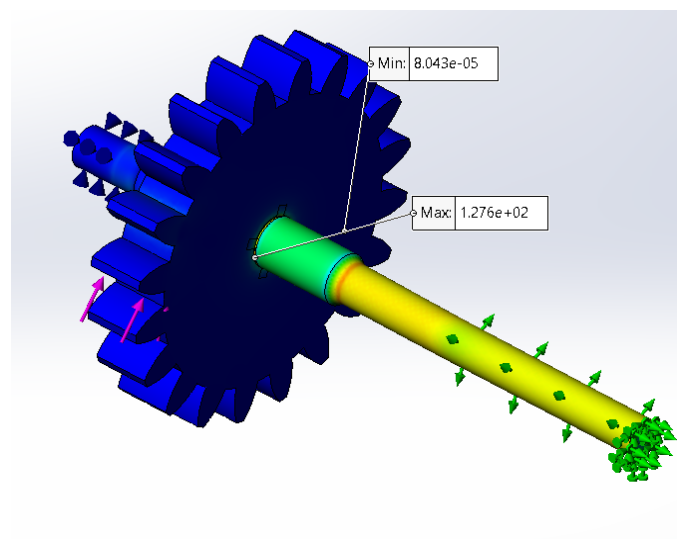


Figure 36: Optimal shaft design simulation tested with pinion load.



Figure 37: Final shaft design

3.3.4 Tolerance

On the shaft there are 3 locations that require specific tolerancing: The motor end of the shaft, the bearing end of the shaft and the splines. Siemens have specified the motor hollow shaft to be of H7 tolerance, and thus to achieve a 'location fit' the shaft must be machined to a h6 tolerance. For the shaft end of the bearing, the SKF recommendation suggested a shaft tolerance of 0 to -10 μm , but did not provide specification of the inner bore tolerance, making it difficult to determine a suitable fit. Ultimately a p6 tolerance was chosen for the shaft, as its tolerance band was only slightly wider, with the same MMC. While this may result in a looser fit the eccentric locking collar on the YET 204 bearing allows for adequate tightening even if the shaft is misaligned. For the splines, the manufacturing of the pinion and shaft manufactured and not parched stock. This means the fit can be adjusted from the hole and shaft bias. To remain consistent with the rest of the part, a h6 tolerance of the splines was chosen, with the pinion bore tolerance being P7, indicative of an interference fit. This was chosen as these two parts require significant torque transmission, without any movement between them. For the cost of materials and machining processes refer to Appendix C.2.1.

3.3.5 Lifetime Expectancy and Servicing

Since solidworks FEA does not categorise the max stress as components of torsional and bending stress, the maximum bending stress will be assumed to be the maximum stress displayed by FEA, while the torsional moment will be taken as 97.78 Nm as found in previous calculations. This is assuming a greater bending stress then in reality, as this maximum value would be a combination of torsional and bending stress.

$$\sigma_{bending} = 127.6 \text{ MPa} = \sigma_{alt}$$

$$\tau_{circlip} = \frac{16T}{\pi d^3} = \frac{16(97.798)}{\pi(0.025^3)} = 36.48 \text{ MPa} = \sigma_{mea}$$

From previous calculations: $S_n = 241.0339 \text{ MPa}$

$$\frac{\sigma_{alt}}{\sigma_{mea}} = \frac{127.6}{36.48} = 3.497$$

Using a goodman diagram:

$$3.497x = -0.3333 + 241.0339$$

$$x = 62.92, y = 220.06$$

$$SF = \frac{587.08}{220.06} = 2.7$$

From the calculations above, the shaft has a SF of approximately 2.7 over its endurance limit for fatigue. To determine the lifetime of the shaft, assume that we limit the number of cycles for the shaft at 10^6 (number of cycles at the endurance limit) , with a continuous load being applied throughout a single trip.

$$N_{max} = 10^6 \text{ cycles}$$

Assuming platform tables at the maximum speed, taking 5 minutes per trip, and 8 hours of constant operation per day:

$$t_{trip} = 5 \text{ min}$$

$$n_{trips} = \frac{8(60)}{5} = 96 \text{ trips per day} = 96 \text{ loads}$$

$$L_{time} = \frac{N_{max}}{n_{trips}} = 10,417 \text{ days} \approx 28.5 \text{ years}$$

This calculated lifetime expectancy is a very rough estimate, as many operating conditions may vary life expectancy. Considering operation on offshore wind turbines, the shafts may be subjected to saline air and most conditions, increasing the likelihood of rust. To ensure proper operating conditions, a servicing of the shafts should occur every 6 months, involving a visual examination for any signs of rust or cracks. The shaft should then be coated with an anti corrosion spray to aid in protection from weather conditions. Shaft alignment checks should also be conducted to ensure no significant deflection has occurred which may alter gear interaction with the teeth of the rack. It is suggested that shafts be replaced every 15 years of operation, ensuring that the fatigue limit is never approached as failure to a shaft may have extremely dangerous consequences.

3.4 Drive Pinions

3.4.1 Hand Calculations

The major design considerations of the pinions is the stress through the teeth during loading and the deflection it causes. Excessive stress through the teeth may cause failure during operation, while excessive deflection will cause teeth meshing issues with the rack. This would alter the performance of the platform and create stability concerns. During the following analysis, force from the rack is assumed to be acting on a single tooth (CR = 1)

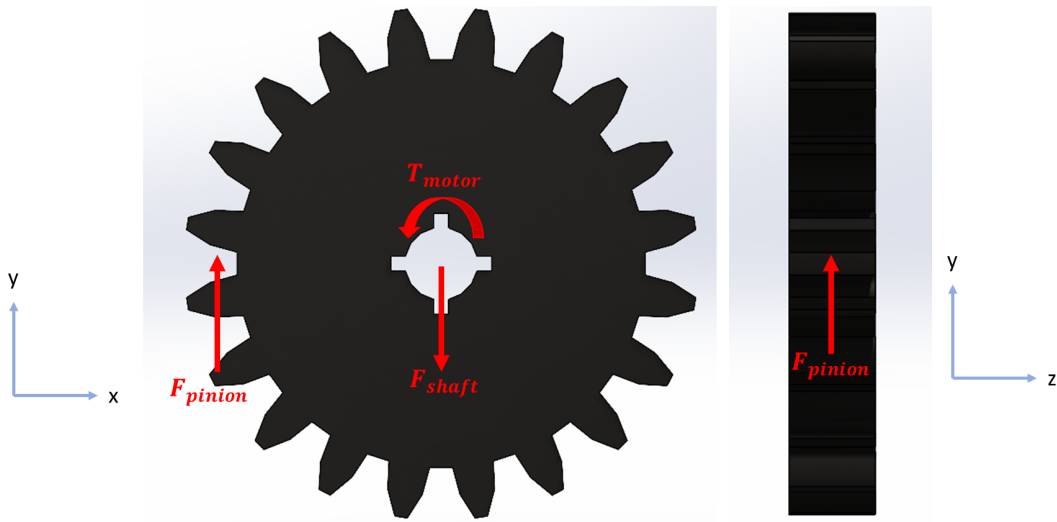


Figure 38: Free body diagram of the drive pinion.

Bending stress on a tooth:

$$\sigma = \frac{F_{pinion}}{mbY}$$

$$m = 8mm, b = 30mm, Y = 0.315$$

$$\sigma = \frac{1222.47}{(0.008)(0.03)(0.315)} = 16.17 \text{ MPa}$$

$$SF = \frac{0.58\sigma_y}{\sigma} = \frac{0.58(620.422)}{16.17} = 22.25$$

3.4.2 FEA

To model the forces acting on the pinion teeth, a fixed geometry fixture was applied to all faces within the bore. As the shaft analysis has already been conducted, the only outcomes of concern are related to tooth stresses and deflection. The force acting on the pinion was 1222 N applied to the face on a single tooth. Note that the deflection being shown in figure 39 is increased by a factor of approximately 5,600.

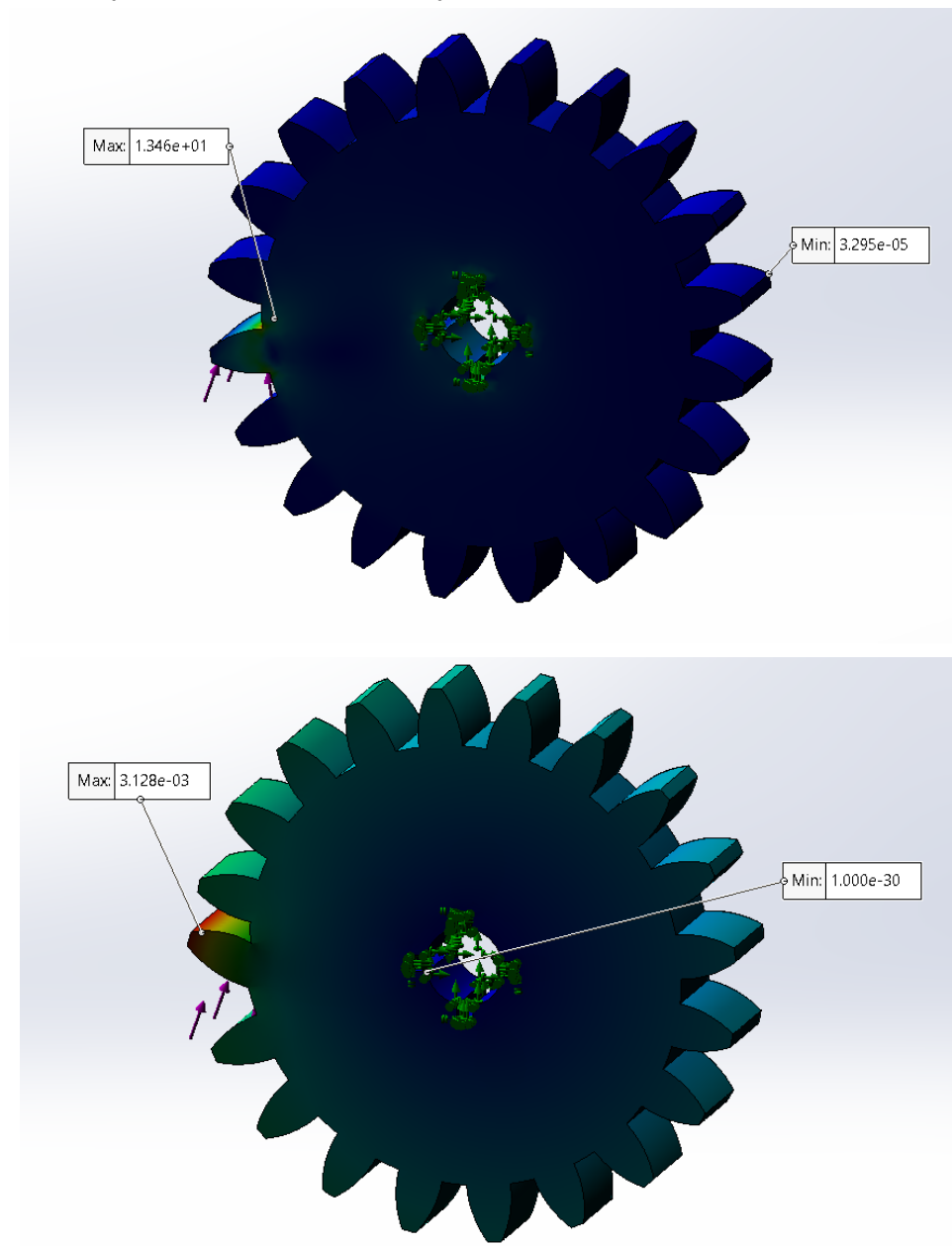


Figure 39: FEA Simulations of the drive pinion showing maximum stress in MPa (top) and deflection in mm (bottom)

From the simulation results, the maximum stress was found to be 13.6 MPa, corresponding to a safety factor of 46. The maximum deflection was found to be 0.003 mm. Whilst both these values are far above any safety thresholds, they are responsible for carrying the entire vertical load of the platform, and it was decided that making the pinions smaller or narrower would provide an insignificant amount of cost reduction and there not with consideration.

3.4.3 Tolerance Calculations

The only tolerancing required is in the bore region, which is required to be H7, which would create an interference transition fit with the splined region of the shaft. An interference fit is required to ensure proper power transmission from the shaft to the drive pinion. For the cost of materials and machining processes refer to Appendix C.2.1.

3.4.4 Lifetime Expectancy and Servicing

Since the loading patterns are similar to the shaft, the drive pinions and shaft should be replaced as an assembly, giving the drive pinions a lifetime of 15 years. Servicing should be conducted every 6 months, involving visual inspection for any cracks or fatigue wear, followed by the application of the anti corrosion spray used during the shaft servicing. Lubrication of the teeth should also be considered and applied if required.

3.5 Shaft Bearings

3.5.1 Hand Calculations

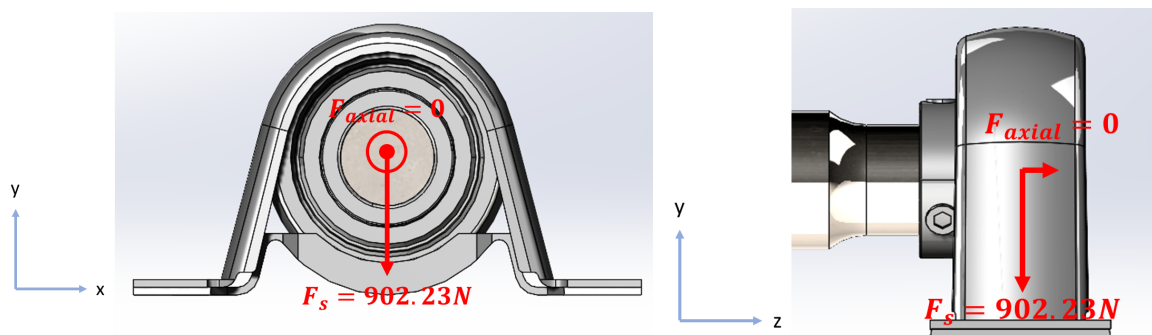


Figure 40: Free body diagram of radial and axial forces acting on the bearing

During this analysis we are assuming that there is no axial load present, light to moderate impact during loading, with 90% reliability.

$$F_a = 0N$$

$$K_a = 1.5$$

$$K_r = 1$$

Design life is also taken to be 30,000 hours, as we are assuming 8 hour daily operation, for all working days.

$$L = 30,000 \text{ hrs}$$

$$N_{\text{bearing}} = N_{\text{shaft}} = 44.12 \text{ rpm; From motor calculations}$$

$$L_{\text{rev}} = 30,000 \times 60 \times 44.12 = 7.9416 \times 10^7 \text{ revolutions}$$

The C value of the selected YET-204 bearings 12.7 kN

From 3.3.1, the calculated radial force on the bearings is 902.28N.

$$F_r = 902.28 \text{ N}$$

To calculate the C min value required for these operating conditions the following formula is applied:

$$C_{\text{req}} = F_r K_a \left(\frac{L_{\text{rev}}}{K_r L_c} \right)^{0.3}$$

$$C_{\text{req}} = (902.28)(1.5) \left(\frac{7.9416 \times 10^7}{(1)(90 \times 10^6)} \right)^{0.3} = 1303.56 \text{ N} \approx 1.3 \text{ kN}$$

For a 20mm diameter shaft, the smallest dynamic C value in the SKF catalogue that fits in the selected pillow block (P-52) is 12.7kN. This is significantly larger than the required value, thus the minimal loading cases on the return trips outlined in the design brief are not considered, as it will not affect bearing choice. For the cost of purchasing the bearing assemblies, refer to Appendix C.2.1.

3.5.2 Lifetime Expectancy and Servicing

To estimate the lifetime of the bearings the following formula is applied:

$$L_{\text{rev}} = K_r L_r \left(\frac{C}{F_r K_a} \right)^{3.33}$$

$$L_{\text{rev}} = (1)(90 \times 10^6) \left(\frac{12,700}{902.28(1.5)} \right)^{3.33} = 1.557 \times 10^{11} \text{ revolutions}$$

$$L_{\text{time}} = \frac{1.557 \times 10^{11}}{44.12} = 3.53 \times 10^9 \text{ minutes} \approx 6713 \text{ years}$$

This is expected considering the significantly larger C value and relatively slow rotation rate of the shaft. Whilst these bearings should not require replacement for the lifetime of the platform, a visual inspection every week should be conducted to ensure that the bearings show no visible signs of fatigue, with a service happening every 6 months. Servicing should consist of removing the bearing from its housing, disassembling, cleaning, lubrication, reassembly and reattachment into its housing. Both the visual inspection and servicing is extremely important, as the bearing may fail via other modes, and thus a proper maintenance schedule should be maintained and bearings replaced at any signs of significant damage.

3.6 Bearing Fasteners

3.6.1 Hand Calculations

To help keep the shaft and pinion stable and secure, bearings are placed on the other side of the pinion. These need to be firmly secured to help prevent any vibration. Due to the symmetry of the system, the force of each shaft onto each bearing will be the same. As such, we can determine whether a selected fastener is appropriate. We know from Section 3.3.2, that the force of the shaft onto the bearing is 902.28 N .

A free body diagram can be seen in Figure 41.

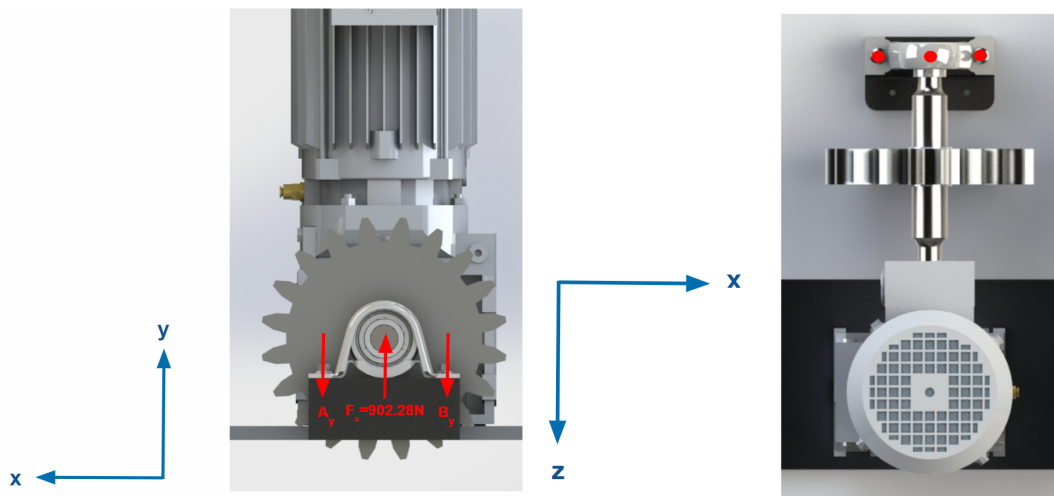


Figure 41: Free Body Diagram of the forces on the bearing.

We can solve for the unknown forces A_y and B_y , as follows:

$$\sum M@A = 0$$

$$-(902.28)(43) + (B_y)(86) = 0$$

$$B_y = \frac{(43)(902.28)}{(86)}$$

$$B_y = 451.14\text{ N}$$

$$\sum F_y = 0$$

$$902.28 - A_y - B_y = 0$$

$$A_y = 902.28 - B_y$$

$$A_y = 902.28 - 451.14$$

$$A_y = 451.14\text{ N}$$

Each bolt will carry the same loading (451.14 N) due to the symmetry of the distribution.

Given that failure of the bolt will be due to tension, we can work out the tensile stress of the bolts. Each bolt has a diameter of 6 mm . Hence,

$$\sigma = \frac{F}{A} = \frac{451.14}{\frac{\pi}{4}(0.006)^2} = 15.96\text{ MPa}$$

The material of the bolt is Alloy Steel, which has a tensile strength of 723.825 MPa and a yield strength of 620.422 MPa . We do not want the material to yield at all, therefore, we can determine the safety factor of each of the bolts to be:

$$SF = \frac{620.422}{15.96} \approx 38.87$$

As this safety factor is very large, the fasteners that hold the bearings will not fail due to tension.

3.6.2 FEA

To accompany the results of the hand calculations and confirm that the fasteners holding the bearings are appropriate and will not yield, FEA was performed on the bearing, bearing block, mount and bolts. The material of the bolt was chosen to be Alloy Steel and it was assumed that the base of the mount will be a fixture and that the loading will occur within the bearing as a point load. Due to it being a curved surface, solidworks distribute this point load among the curved surface, which is a reasonable assumption as the load would be distributed and not occur just in the centre. Furthermore, it was assumed that the bearing itself was rigid and that the housing is what can deform, which is also a reasonable assumption as the loading will be distributed throughout the housing to the bolt and it is the bolt that may cause it to deform slightly.

Figure 42 illustrates the Von Mises Plot of the fasteners.

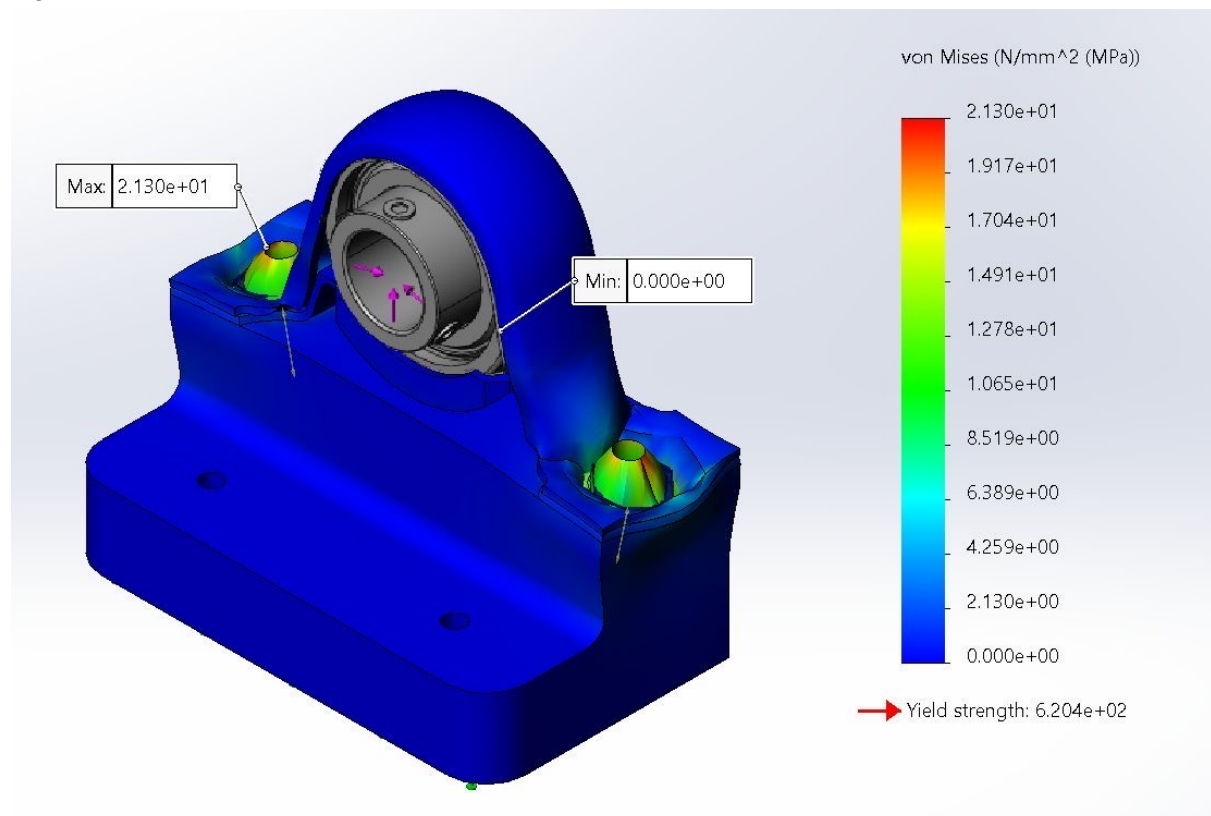


Figure 42: Von Mises Plot of Bearing Fasteners.

As can be seen from Figure 42, a lot of the stress will be occurring in the top part of each bolt. This makes sense as the forces generated by the bearing to oppose that of the shaft is upwards, meaning that it will try to lift the bearing block off the mount and as such exert force upwards onto the heads of the fasteners.

It can also be seen from the Figure 42 that the maximum stress is: 21.3 MPa. While this is not the same force as calculated by hand, it is reasonably close and is logical to see this

slight deviation due to assumptions made in the hand calculations, such as a single point load located in the centre of the bearing which in fact is not realistic.

Since the material of the bolt is Alloy Steel, which has a yield strength of 620.422 MPa, we can determine the safety factor of each of the bolts to be:

$$SF = \frac{620.422}{21.3} \approx 29.13$$

As this safety factor is still reasonably large, we can conclude that the M6 fasteners that hold the bearings will not fail due to tension and can, therefore, be used.

3.6.3 Servicing

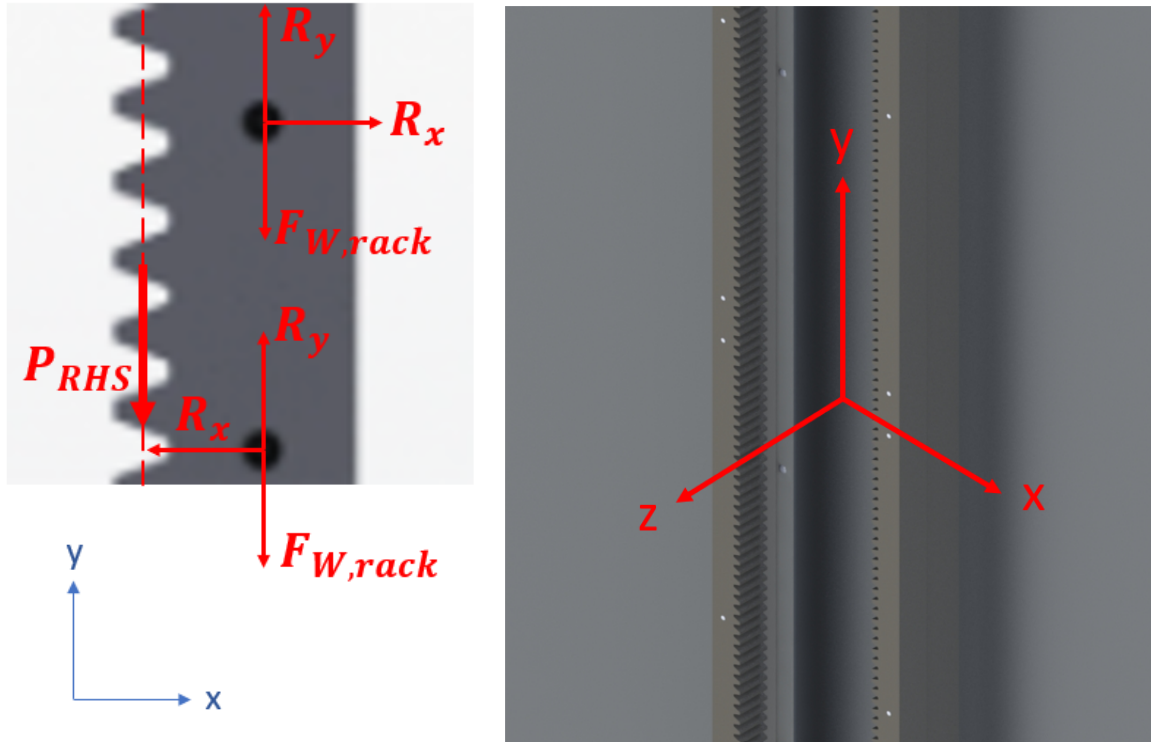
In theory, the fasteners that hold the bearing should not need to be replaced, however, it would be good to check them and see if they require replacing or servicing once every month. Given that a lot of wind turbines are along or in the ocean, there is going to be a lot of moisture and so rust is very common and more likely to occur. Rust is a type of corrosion which can contribute to the loosening of fasteners and can initiate fatigue cracks in the threads. As such it is important to make sure that the bolts do not get rusted.

To prevent rust, everytime new bolts are placed on, protective coatings and sealants should be used. Additionally, purchasing specially formulated, rust resistant fasteners is also advised. A zinc/nickel coating is expected to last at least 1,000 hours (which is approximately 42 days) before the bolt starts to rust [2] and as such, the bolts should be looked at and a new coat should be applied once every month. If this process has become neglected and bolts begin to rust, it is important that they are removed and replaced immediately.

3.7 Mounting system

3.7.1 Hand Calculations

3.7.1.1 Vertical Rack to Channel Beam Calculations



Figures 43: The FBD of the rack to beam mounting system

Assumptions

- All reaction forces in the x-z plane are through the guiding wheels
- Vertical force from pinions acts directly on the centre of the rack
- The weight of the rack acts in line with the fasteners, each fastener supports one third of each rack of 1506.45 mm [Appendix B.4]
- The maximum moment from the pinion force occurs when acting halfway between two fasteners 100 mm apart, as pinion force can act at any vertical point
- Given the symmetry of the climbing mechanism and platform, the load
- From previous calculations the force exerted by each pinion on the rack is 1222.47315 N and the force exerted by each guide wheel is 1339.38 N

Vertical reaction force:

$$F_{w,rack} = 44 \text{ kg/m} * 0.50215 \text{ m} * 9.81 \text{ m/s}^2$$

$$P_{RHS} = 1222.47315 \text{ N}$$

$$\Sigma F_y = 0 = 3 * R_y - 3 * F_{w,rack} - P_{RHS}$$

$$R_y = 624.239076 \text{ N}$$

Horizontal reaction force:

$$\Sigma M_{@centre \text{ between rivets}} = 0 = 41 * P_{RHS} - 2 * 50 * R_x$$

$$R_x = 501.2139915 \text{ N}$$

Maximum shear stress on bolt:

$$R_{max} = \sqrt{R_x^2 + R_y^2} = 800.5559876 \text{ N}$$

Given that failure will occur first as shear failure in the 6mm diameter fasteners,

$$\tau = \frac{R_{max}}{A_{rivet}} = \frac{800.556}{\pi * 0.003^2} = 28313876.14 \text{ Pa} \approx 28.3 \text{ MPa}$$

Given the material is Alloy Steel, it has an Ultimate Shear Stress (USS) of 542.85 MPa, the safety factor is 19.175.

3.7.1.2 Channel Beam to Tower Calculations

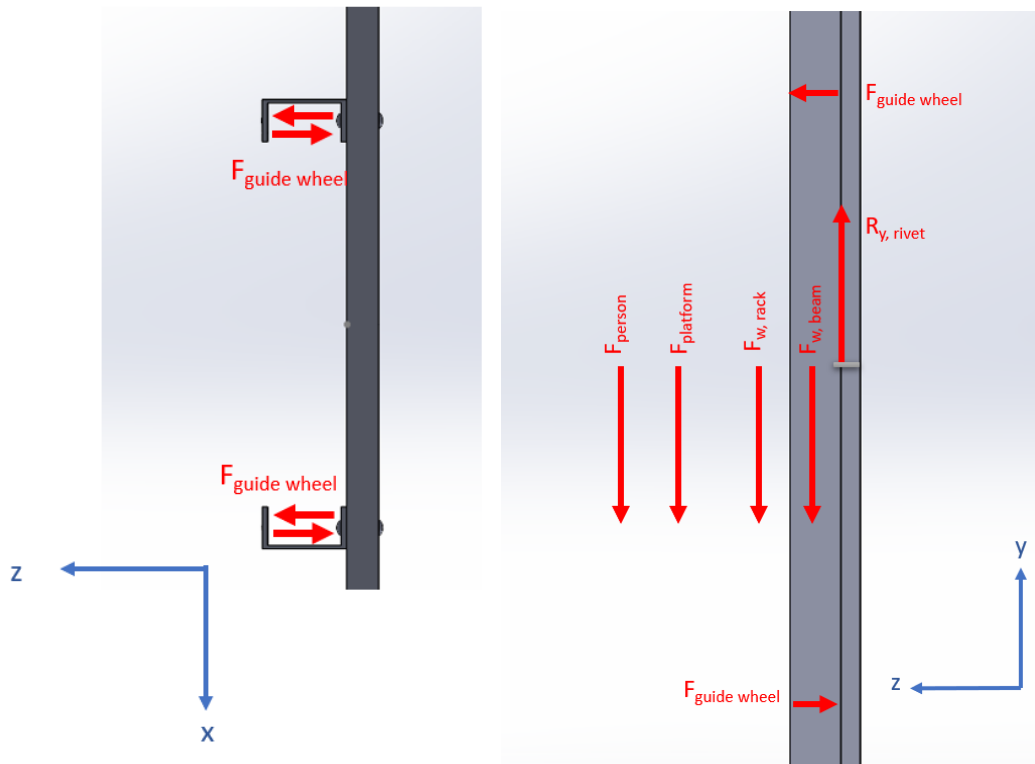


Figure 44: Channel beam to tower attachment FBDs

Assumptions

- For vertical (y) reaction force, max load when platform load is acting on a single fastener
- As fasteners are spaced 1 m apart, each supports 1m of beam and 1m of rack
- X-plane guide wheel reaction forces are equal and account for any reaction forces in their plane.
- Platform and cargo loads are symmetrical and centred, thus each side of the mounting takes exactly half the load

- Platform load is calculated 3.11

$$\begin{aligned}
 F_{w,rack} &= 44 \text{ kg/m} * 1 \text{ m} * 9.81 \text{ m/s}^2 \\
 F_{w,beam} &= 8.33 \text{ kg/m} * 1 \text{ m} * 9.81 \text{ m/s}^2 \\
 F_{platform} &= 487.13 \text{ N} * 0.5 \\
 F_{person} &= 120 \text{ kg} * 0.5 * 9.81 \text{ m/s}^2 \\
 \Sigma F_y = 0 &= R_{y,rivet} - F_{w,rack} - F_{w,beam} - F_{platform} - F_{person} \\
 R_y &= 1345.5223 \text{ N}
 \end{aligned}$$

Assuming failure due to shear stress in the fasteners, which have a diameter of 10mm:

$$\tau = \frac{R_{max}}{A_{rivet}} = \frac{1345.5223}{\pi * 0.005^2} = 17131722 \text{ Pa} \approx 17.13 \text{ MPa}$$

Given the material is Alloy Steel, it has an Ultimate Shear Stress (USS) of 542.85 MPa, the safety factor is 31.69.

3.7.2 FEA

To support the hand calculations, finite element analysis of the mounting system was conducted. A von Mises stress analysis demonstrates whether the system will buckle or yield under the applied loads, and a displacement analysis demonstrates whether there would be significant deformation under the applied loads.

The fasteners of the system were simulated using the solidworks Toolbox to simplify the analysis process, and the diameter and material remained the same. The maximum possible load was simulated by applying the load force from each pinion onto the single face of a rack tooth, on each side, normal to its face. This is indicated by the pink arrows in the images below. The tower wall is evaluated as a fixed feature, as it was indicated in the design brief that it can withstand any loads from the entire climbing mechanism with no laying effects.

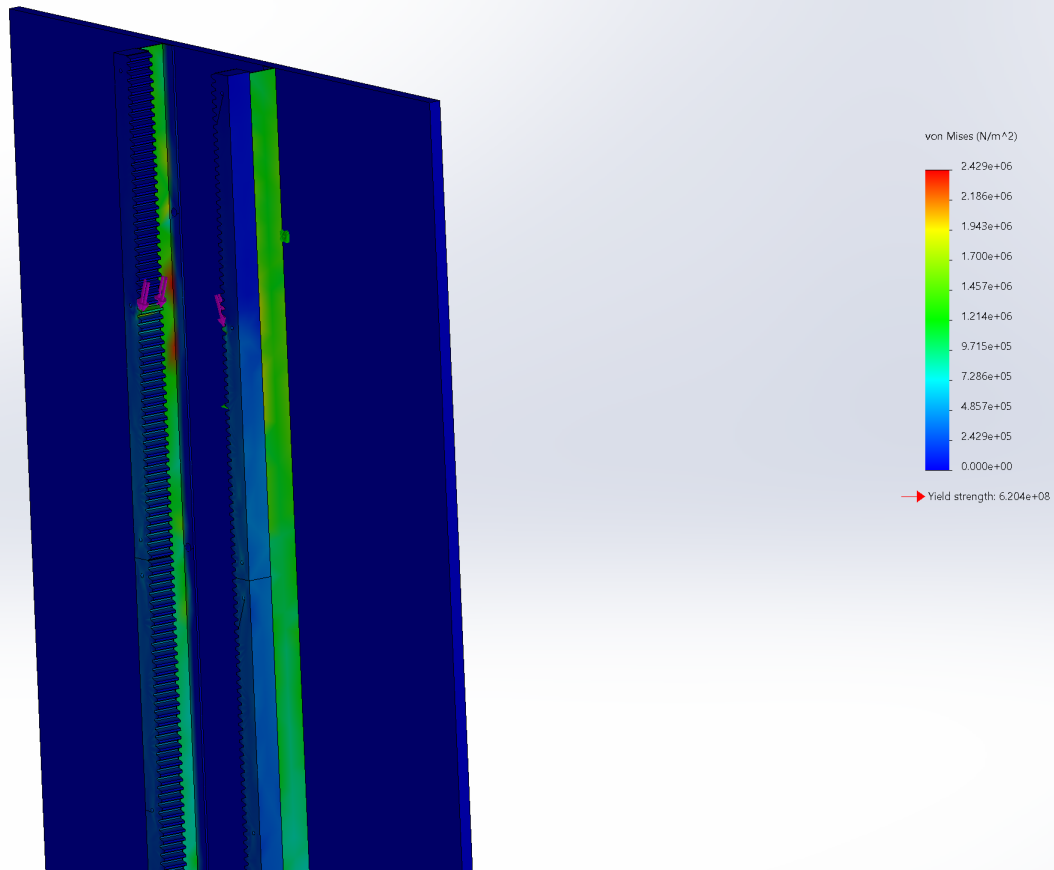


Figure 45: The von Mises stress analysis of the mounting system for a length of 12m. The system reaches a maximum of 2.429 MPa, in the supporting channel beam, and has a yield strength of 620.4 MPa, resulting in a safety factor of >255. As such the system will not fail due to stress.

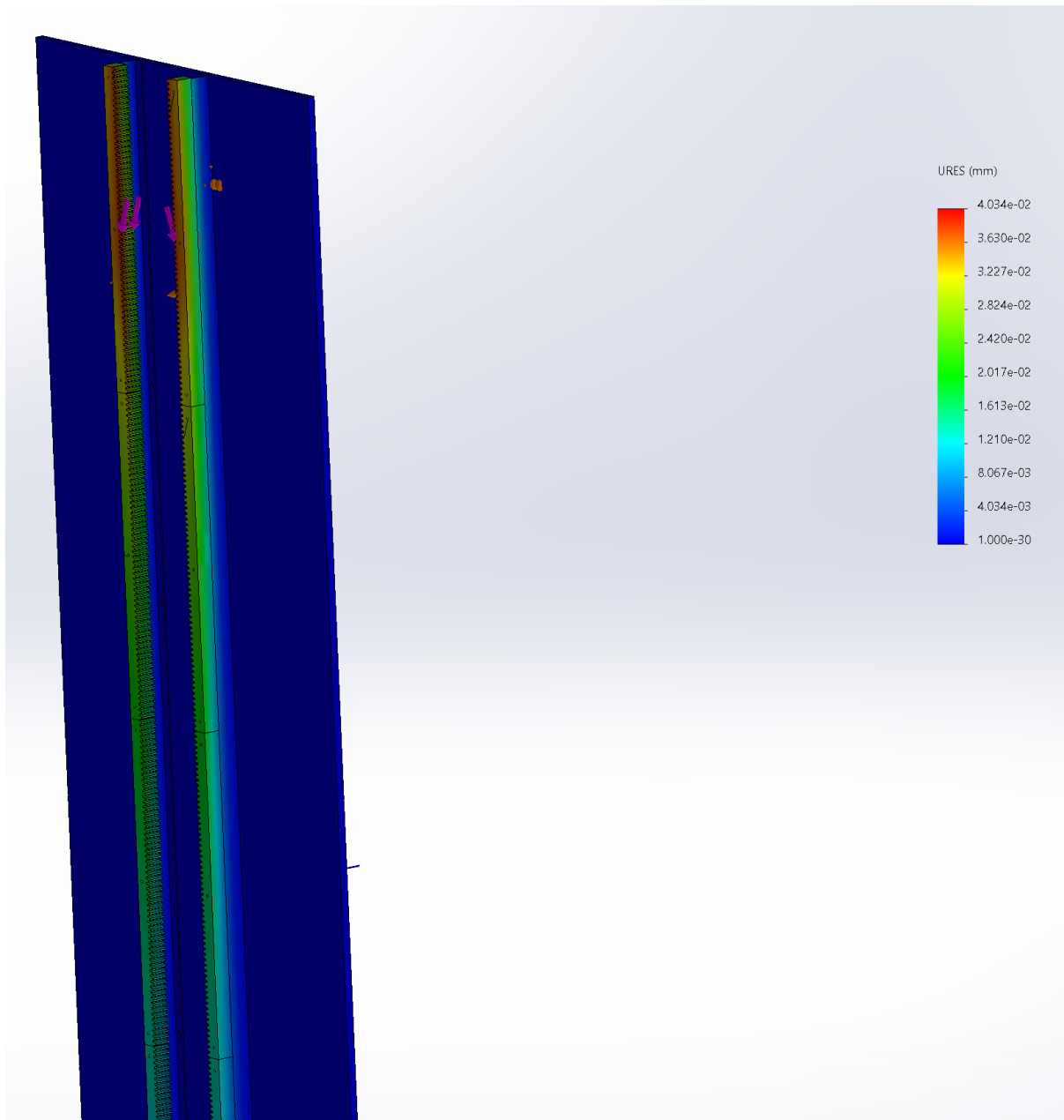


Figure 46: The displacement analysis for the mounting system over 12 metres. This analysis indicates that the greatest displacement will occur around the teeth where the load is applied, focused on the faces furthest from the tower wall. This displacement reaches a maximum of 0.04034 mm.

3.7.3 Lifetime Expectancy and Servicing

From the rack datasheets, they have a strength endurance of 10,000 hours and are made of bright steel, which is highly durable and has some corrosion resistance. As such the racks need only be inspected for damage such as cracks and tooth wear approximately every 100 uses. The steel alloy channel/parallel flange beams are also corrosion resistant and durable, with a grade of 300 MPa [Appendix B.4]. Given the maximum experienced stress of 2.429 MPa is experienced, the beams can also be inspected along the same timeline as the racks.

The stainless steel fasteners are also coated, making them non-corrosive even in marine and coastal environments.

It is not expected that any part of the mounting system will require replacement within the first 20 years of use, given the infrequency of the system use overall and their corrosion resistance.

3.8 Scissor Lift

3.8.1 Purchase and Specifications

While a feasible scissor lift was designed, as mentioned in Section 2.2.3, due to the short timeframe for this project, we believe that the design of a scissor lift is out of the scope of this project as it could be a project on its own. As such, instead of making a scissor lift, for now, a scissor lift will be purchased. We require a mobile scissor lift that can hold the platform and lift it up to a maximum height of 1 metre.

After investigating multiple options, the HW1002 from Jinan PLK Machinery Co., Ltd was chosen and can be seen in Figure 47. The scissor lift catalogue and cost can be found in Appendix B.3 and Appendix C.2.3, respectively.



Figure 47: Mobile Scissor Lift [3]

The HW1002 model is able to hold a load of 1000 kg, which is enough to withhold the current weight of the platform with a person with a safety factor of 3. Additionally, it is able to lift the platform to a maximum height of 1m which is what is required. Furthermore, the dimensions of the top are 1.6m by 1.0m, which is large enough to hold our platform. Furthermore, it has a reasonably fast lifting time of approximately 20 seconds and has wheels (that can lock in place) which make it mobile and easier for a single person to move. As it satisfies all the requirements, this scissor lift should be purchased.

3.8.2 Lifetime Expectancy and Servicing

According to the Australian standard for scissor lifts, AS 2250.10-2006, scissor lifts must undergo a major inspection and servicing every ten years from their date of manufacture [4]. Additionally, if appropriately used and performing regular maintenance, a scissor lift can last around 30 years [5]. As such, due to this long lifetime expectancy, it seems to be a good investment decision.

3.9 Safety Brace

3.9.1 Hand Calculations

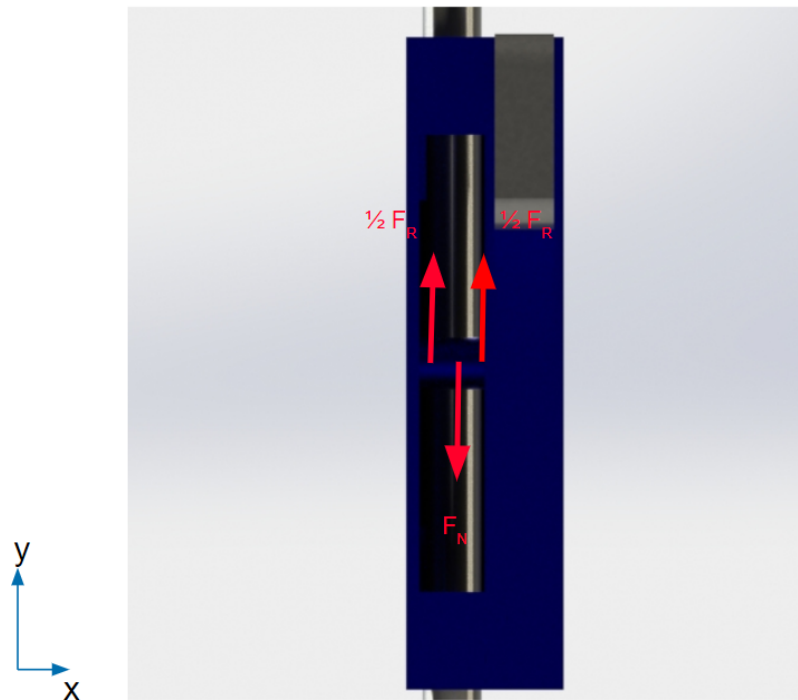


Figure 48: Beam supporting safety brake mechanism

In order to calculate the strength of the safety case, it is reasonable to choose the beam that supports the braking mechanism, since this component is most likely to fail. We assume that the entire load is transferred to this beam and that the mass of the beam and brake that rest on the beam is negligible, due to their small size and thus, small weight.

$$\begin{aligned}F_N &= m_{\text{load}} \cdot a_{\text{load}} \\F_N &= 100\text{kg} \cdot 9.81\text{m/s}^2 = 981\text{N} \\L_{\text{beam}} &= 14\text{mm} = 0.014\text{m} \\d &= 8\text{mm}\end{aligned}$$

$$\begin{aligned}\sum F_y &= 0: \\-F_N + \frac{1}{2} F_R + \frac{1}{2} F_R &= 0 \\F_R &= F_N = 981\text{N}\end{aligned}$$

$$M = F_N \cdot L/4 = 981 \cdot 0.014/4 = 3.43\text{Nm}$$

$$\sigma_a = F/A = 981/(\pi(8)^2/4) = 19.5 \text{ MPa}$$

$$\sigma_b = Mc/I = 32M/(\pi d^3), \text{ for small, solid cylindrical beam}$$

$$\sigma_b = 32 \cdot 3.43/(\pi \cdot 0.008^3) = 68.3 \text{ MPa}$$

$$\tau = Tr/J = 0 \text{ MPa, since no torque}$$

Selection of material is alloy steel:

$$S_y = 620.4 \text{ MPa}, S'_n = 0.5S_y = 310.2 \text{ MPa}$$

$$S_n = S'_n C_L C_G C_S C_T C_R$$

$$S_n = 310.2 \cdot 1 \cdot 0.9 \cdot 0.9 \cdot 1 \cdot 1 = 251.26 \text{ MPa}$$

$$\sigma_{ea} = \sigma_b = 68.3 \text{ MPa}$$

$$\sigma_{em} = \sigma_a/2 + \sqrt{\tau^2 + (\sigma_a/2)^2} = 19.5 \text{ MPa}$$

Goodman Diagram lines:

1. $y = 68.3/19.5 \cdot n$
2. $y = -251.26/620.4 \cdot n + 251.26$

Solving for n and y:

$$n = 64.3 \text{ MPa}$$

$$y = 225 \text{ MPa}$$

$$SF = 64.3/19.5 = 225/68.3 = 3.3$$

Thus the beam will not fail under the load.

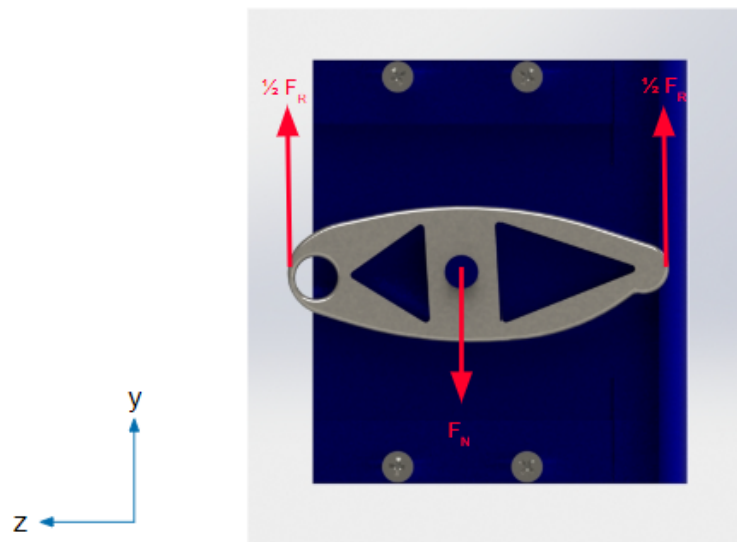


Figure 49: Safety brake mechanism

In order to calculate the stress on the beam, assume the beam has a solid square cross-section and forces act through the central line of the beam. Also assume the weight of the brake is negligible due to small mass.

$$F_N = m_{load} \cdot a_{load}$$

$$F_N = 100 \text{ kg} \cdot 9.81 \text{ m/s}^2 = 981 \text{ N}$$

$$L_{\text{beam}} = 80\text{mm} = 0.080\text{m}$$

To utilise the stress concentration table:

$$b=40\text{mm}$$

$$h=10\text{mm}$$

$$d= 8\text{mm}$$

$$\begin{aligned}\sum F_y &= 0: \\ -F_N + \frac{1}{2} F_R + \frac{1}{2} F_R &= 0 \\ F_R &= F_N = 981\text{N}\end{aligned}$$

$$\begin{aligned}M &= F_N * L / 4 = 981 * 80 / 4 = 19620\text{Nmm} \\ \sigma_a &= F / ((b-d)h) = 981 / ((40-8)*10) = 3.07\text{MPa} \\ \sigma_b &= Mc / I = 6M / ((b-d)h^2) \\ \sigma_b &= 19260 * 6 / ((0.04-0.008)*0.01^2) = 36.8\text{MPa} \\ \tau &= Tr / J = 0\text{MPa, since no torque}\end{aligned}$$

Selection of material is alloy steel:

$$S_y = 620.4\text{MPa}, S'_n = 0.5S_y = 310.2\text{MPa}$$

$$\begin{aligned}S_n &= S'_n C_L C_G C_S C_T C_R \\ S_n &= 310.2 * 1 * 0.9 * 0.9 * 1 * 1 = 251.26\text{MPa}\end{aligned}$$

Plate with central hole Axial Graph:

$$d/b = 8/40 = 0.2, K_{t,a} = 2.45$$

Bending Graph:

$$d/b = 0.2, d/h = 10/10 = 1, K_{t,b} = 1.9$$

$$\begin{aligned}\sigma_{ea} &= \sigma_b * K_{t,b} = 69.9\text{MPa} \\ \sigma_{em} &= (\sigma_a/2 + \sqrt{\tau^2 + (\sigma_a/2)^2}) * K_{t,a} = 7.52\text{MPa}\end{aligned}$$

Goodman Diagram lines:

1. $y = 69.9/7.52 * n$
2. $y = -251.26/620.4 * n + 251.26$

Solving for n and y:

$$n = 25.8\text{MPa}$$

$$y = 240\text{MPa}$$

$$\text{SF} = 25.8/7.52 = 240/69.9 = 3.43$$

Thus the brake will not fail under the load.

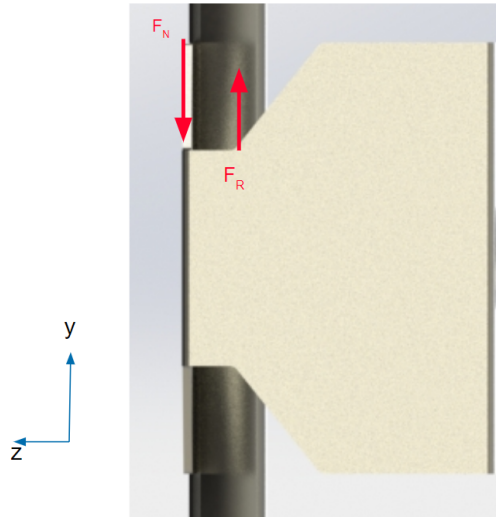


Figure 50: Cable Mounting Bracket

Assume for analysis that all of the load is transferred to the smaller areas of the bracket, since these parts are likely to fail first. Also, neglect the weight of the bracket due to minimal mass.

Note, that there are 2 mounting brackets, thus, assume each bracket shares load equally.

$$F_N = \frac{1}{2} * m_{load} * a_{load}$$

$$F_N = \frac{1}{2} * 100\text{kg} * 9.81\text{m/s}^2 = 490.5\text{N}$$

$$L_{beam} = 5\text{mm} = 0.005\text{m}$$

$$\sum F_y = 0:$$

$$-F_N + F_R = 0$$

$$F_R = F_N = 490.5\text{N}$$

$$\sum M = 0: M - F_N * 0.005 = 0$$

$$M = 2.4525\text{Nm}$$

$$\sigma_a = F/A = 490.5/(1*25) = 19.62\text{MPa}$$

$$\sigma_b = Mc/I, I = hb^3/12 \text{ and } c = \frac{1}{2}b, \text{ where } b=25\text{mm}, h=1\text{mm}$$

$$\sigma_b = 2.4525 * 0.0125 / (1.3 * 10^{-9}) = 30.66\text{MPa}$$

$$\tau = Tr/J = 0\text{MPa, since no torque}$$

Election of material is alloy steel:

$$S_y = 620.4\text{MPa}, S'_n = 0.5S_y = 310.2\text{MPa}$$

$$S_n = S'_n C_L C_G C_S C_T C_R$$

$$S_n = 310.2 * 1 * 0.9 * 0.9 * 1 * 1 = 251.26\text{MPa}$$

$$\sigma_{ea} = \sigma_b = 30.66\text{MPa}$$

$$\sigma_{em} = \sigma_a/2 + \sqrt{\tau^2 + (\sigma_a/2)^2} = 19.62\text{MPa}$$

Goodman Diagram lines:

1. $y = 30.66 / 19.62 * n$
2. $y = -251.26 / 620.4 * n + 251.26$

Solving for n and y:

$$n = 127.7 \text{ MPa}$$

$$y = 199.5 \text{ MPa}$$

$$\text{SF} = 127.7 / 19.62 = 199.5 / 30.66 = 6.5$$

Thus the bracket will not fail under the load.

3.9.2 FEA and Optimisation

In order to complement the hand calculations of the safety brake mechanism a finite element analysis was undertaken in order to ensure that the mechanism won't fail when being used. In order to optimise the braking mechanism, FEA was used to produce better designs that cut down on overall weight of the part and thus the cost needed for manufacturing.

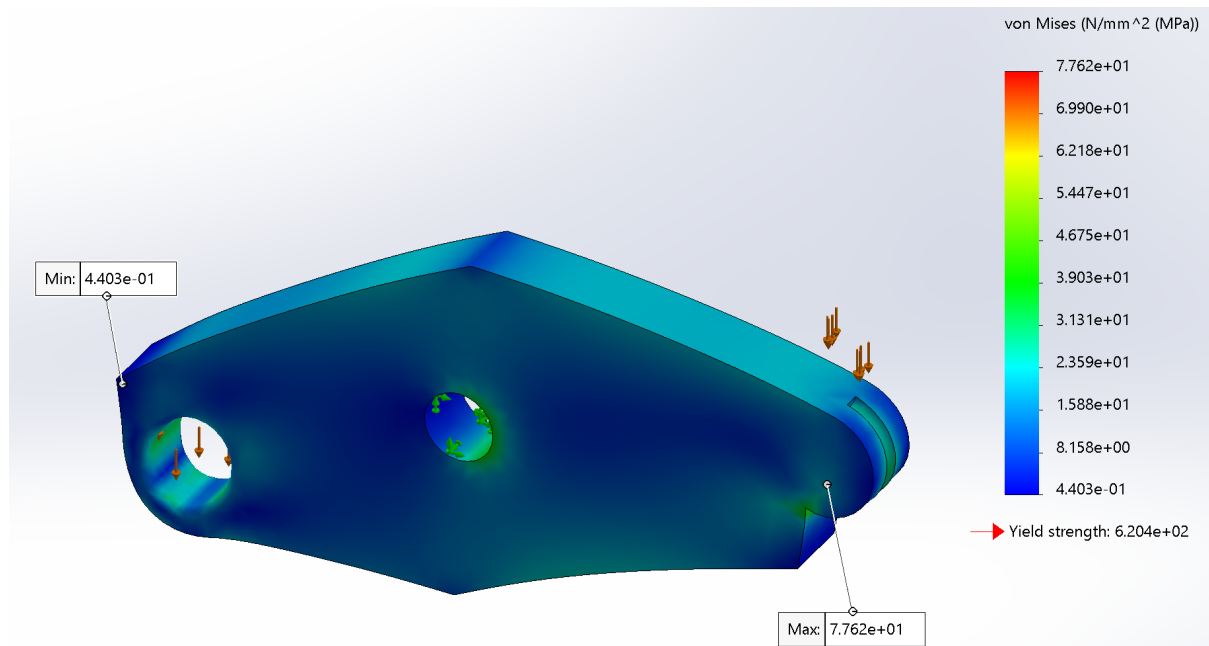


Figure 51: Design 1 of the safety brake mechanism.

For the first design of the part, it can be seen from the Von Mises plot shown in the figure above, that the maximum stress on the mechanism is 77.62 MPa. Since, the material selected for the mechanism is Alloy Steel, the Yield Strength is 620.4 MPa and thus the safety factor of the part is 7.99.

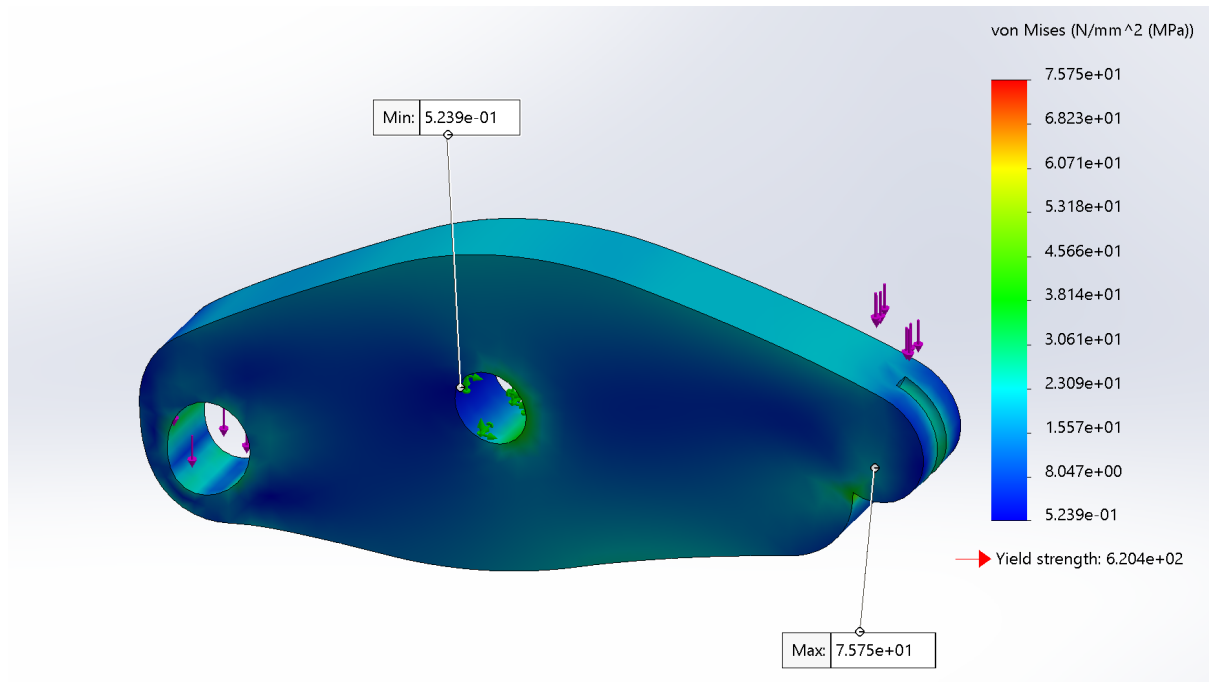


Figure 52: Design 2 of the safety brake mechanism.

For design 2 of the safety brake, the part was optimised by using a 10 mm fillet on all the sharp edges of the part, excluding the outside edges. This produced a Von Mises plot with a maximum stress of 75.75 MPa, since the Yield Strength of the part is 620.4 MPa this produces a safety factor of 8.2.

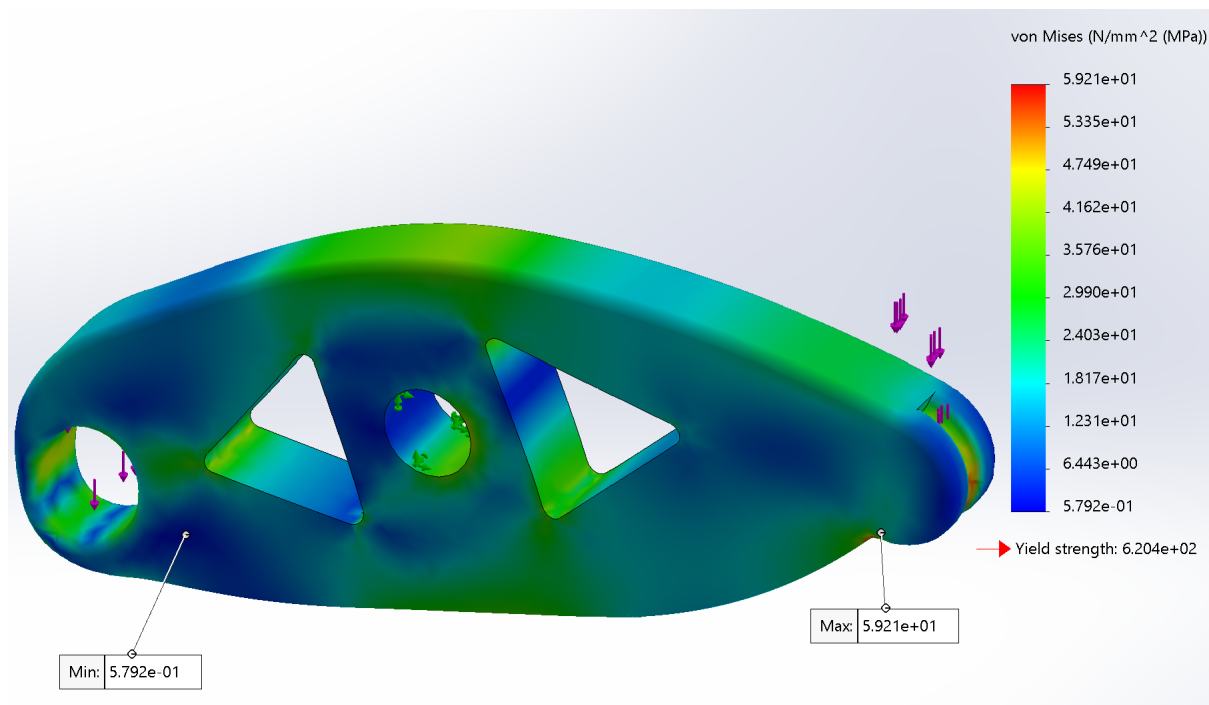


Figure 53: Design 3 of the safety brake mechanism.

For design 3 of the safety brake, the part was optimised by using a 50 mm fillet on all the sharp edges of the part, excluding the outside edges, which had a 1 mm fillet applied to

them. Also, to cut down on the weight of the part 2 triangles were cut out of the main body. This produced a Von Mises plot with a maximum stress of 59.21 MPa, since the Yield Strength of the part is 620.4 MPa this produces a safety factor of 10.48. This not only decreased the weight and thus manufacturing price, but also strengthened the part.

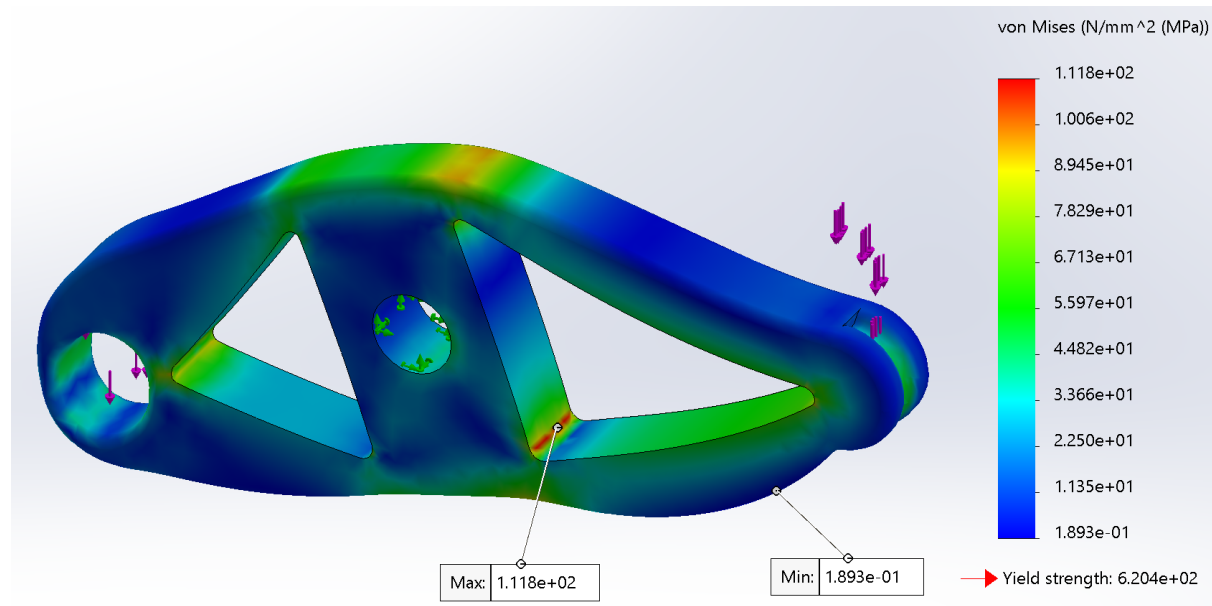


Figure 54: Design 4 of the safety brake mechanism.

For design 4 of the safety brake, the part was optimised further by making the triangular cut outs bigger. This produced a Von Mises plot with a maximum stress of 111.8 MPa, since the Yield Strength of the part is 620.4 MPa this produces a safety factor of 5.55. Although this decreased the weight and manufacturing price further, it did however decrease the strength of the part. Since, the safety factor of the part is still greater than 3, which was the agreed upon safety factor for the whole design, the part still functions without failing and thus, design 4 is the optimal design for the part.

3.9.3 Lifetime Expectancy and Servicing

Since, the part will be used right by the ocean, it is subject to being exposed to seawater. Because the material of the part is Alloy Steel, which is not naturally corrosion-resistant, the part should be inspected at least once every two months.

The part is also designed to be used once, as it is supposed to prevent injury or death in case the lift mechanism fails, thus the safety mechanism should be replaced after being used for its proper function. For costing information, refer to Appendix C.2.2.

4 Conclusion

The final design for the wind turbine tower climbing mechanism is a versatile system with an emphasis on a long lifetime expectancy, portability of the climbing platforms and the serviceability of the components. The core components of the system include the combined operator and cargo platform, the dual motor and pinion system, tower mounting system and the customised safety mechanisms. A scissor lift is also utilised to transport the platform between turbines, which allows for precise initial positioning of the pinions on the vertical racks.

Key features of the platform which maximise its quality include the foldable safety rail for ease of transfer on and off the platform, the dead man's switch and guide wheels which maintain the platform's centred position between the racks as well as remaining parallel vertically to the racks. The climbing mechanism consists of two pinions, each driven by an rpm reduced motor with helical worm gears to prevent back driving, interacting with vertical racks mounted to the turbine tower wall. The customised safety mechanism was optimised to reduce mass and maintain simplicity for use.

The design meets all design requirements, with the cost for the system accounted for and minimised as possible in the appendices, redundancy measures in place, and the capability of climbing the 120 metres in the required time. The total mass of the climbing mechanism, not including the tower mounting, is 129.23 kg, and the total cost for one system is \$16,093.85, excluding the vertical racks.

5 References

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